Structural Equation Modeling with Onyx

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Invalid Date

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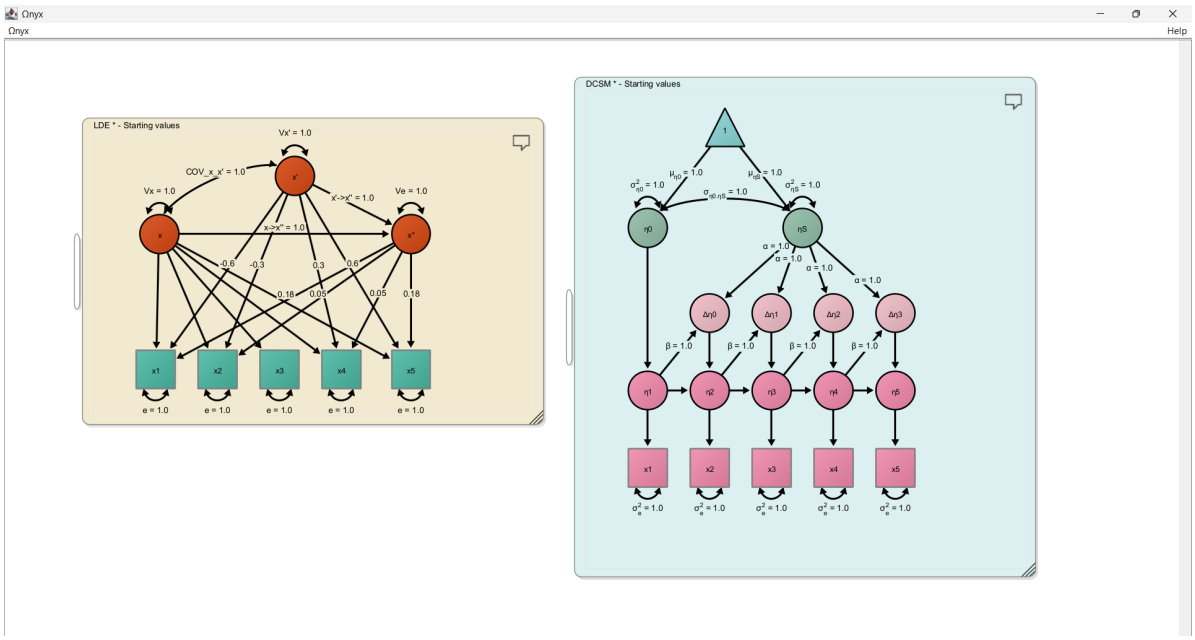
Preface

This is a self-paced workshop for Onyx, a graphical user-interface for structural equation modeling.



This workshop is designed for researchers and students who are new to Onyx and already familiar with basic SEM concepts.

We assume that you are already familiar with:

- latent and observed variables,
- factor loadings and residual variances,
- model fit indices such as chi-square, RMSEA, CFI, and SRMR.

What you will learn

The workshop covers practical Onyx workflows, including:

- creating path models,

- estimating and evaluating model fit,
- comparing nested models,
- working with multigroup models,
- exporting syntax to other SEM tools.

Before you begin

Please make sure that:

1. Onyx is installed and opens normally on your machine.
2. You can load workshop data files from the `sem-workshop/data/` folder.
3. You are comfortable using right-click context menus (most Onyx actions are menu-driven).

If you are new to Onyx, start with **Introduction** and then **Basic Modeling** before moving on to the other chapte The workshop has some introductory chapters that cover basic workflow elements in Onyx, such as basics of model specification, evaluating model fit, or specifying multiple group models.

1 Introduction

1.1 What is Onyx?

Ωnyx is a free software environment for creating and estimating structural equation models (SEM). It provides a graphical user interface that facilitates an intuitive creation of models, and a powerful back end for performing maximum likelihood estimation of parameters. Path diagrams in Ωnyx can be exported to OpenMx, lavaan, and Mplus to allow an easy transition to script-based SEM software. Also, OpenMx scripts can be transformed into a path diagrams in Ωnyx.

1.2 Installation

If you have not worked with Onyx before, make sure to download and install it first. You find some instructions under **Installation**.

1.3 The Interface

Onyx uses a highly interactive graphical workflow. When you open the program, you begin with an empty desktop and build models step by step. Most actions are available through context menus, right-click drag operations, and keyboard modifiers.

1.4 Start here (first-time users)

If this is your first time using Onyx, follow this order:

Basic Modeling — create variables and paths, free parameters, and style diagrams. 2. **Model Fit** — inspect fit indices and residual diagnostics. 3. **Model Comparison** — compare nested models. 4. **Multi-Group Models** and **Factor Scores** — move to more advanced workflows. 5. Exercise chapters — apply the workflow to full examples.

1.5 Interaction basics

Onyx interactions depend on where you click and whether modifier keys are pressed.

- **Double-click on empty model space**: create a latent variable.
- **Shift + double-click on empty model space**: create an observed variable.
- **Right-click drag from one variable to another**: create a directed path.
- **Shift + right-click drag**: create a covariance path.
- **Right-click an object**: open object-specific options.

1.6 A quick first check

Before starting the workshop, try this mini-task:

1. Create an empty model.
2. Add one latent and one observed variable.
3. Draw a path from latent to observed.
4. Free the path parameter.

If this works, your setup is ready and you can proceed with the chapters.

This workshop is still being refined. Chapters include conceptual background, practical demonstrations, and exercises to follow along. There is no need to follow the order of the modules as they are presented. Pick a topic about which you would like to learn more. The topics contain some basic information, some demonstrations, and practical exercises for you to follow along.

1.7 What is expected background knowledge

You will get the most out of this material if you are comfortable with:

- Latent vs. observed variables.
- Factor loadings and residual variances.
- Identification constraints.
- Basic model fit indices (e.g., chi-square, CFI, RMSEA, SRMR).

If these terms are unfamiliar, review a short SEM primer first and then return to this workshop.

2 What you need before starting

- A working installation of **Onyx** on your computer.
- A mouse or trackpad that supports right-click actions (many Onyx actions use context menus). We definitely recommend using an external mouse as input device.
- The workshop data files in `sem-workshop/data/` or online https://github.com/brandmaier/lip2026-onyx-workshop/tree/main/sem-workshop/data.
- Basic familiarity with SEM terms such as latent variables, loadings, residual variances, model fit, and nested model comparison.

2.1 Installation

You can download the most recent version of Ωnyx for all operating systems here (including Windows, Mac OSX, and Unix/Linux here).

2.2 Prerequistes

Ωnyx requires a JAVA runtime environment. If you haven't installed JAVA, please download a recent version, for example OpenJDK (https://openjdk.org/index.html).

2.3 Download

For this workshop make sure to install the *Developer version / unstable*, which contains the latest features:

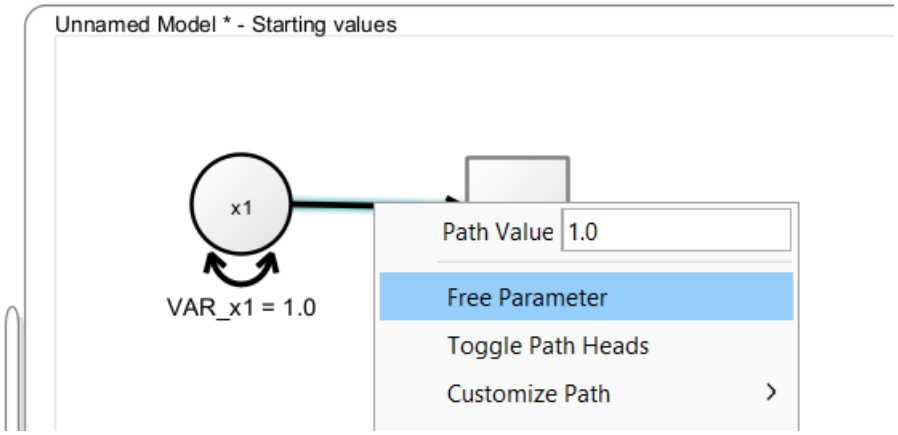
https://onyx-sem.com/download/

3 Basic Modeling

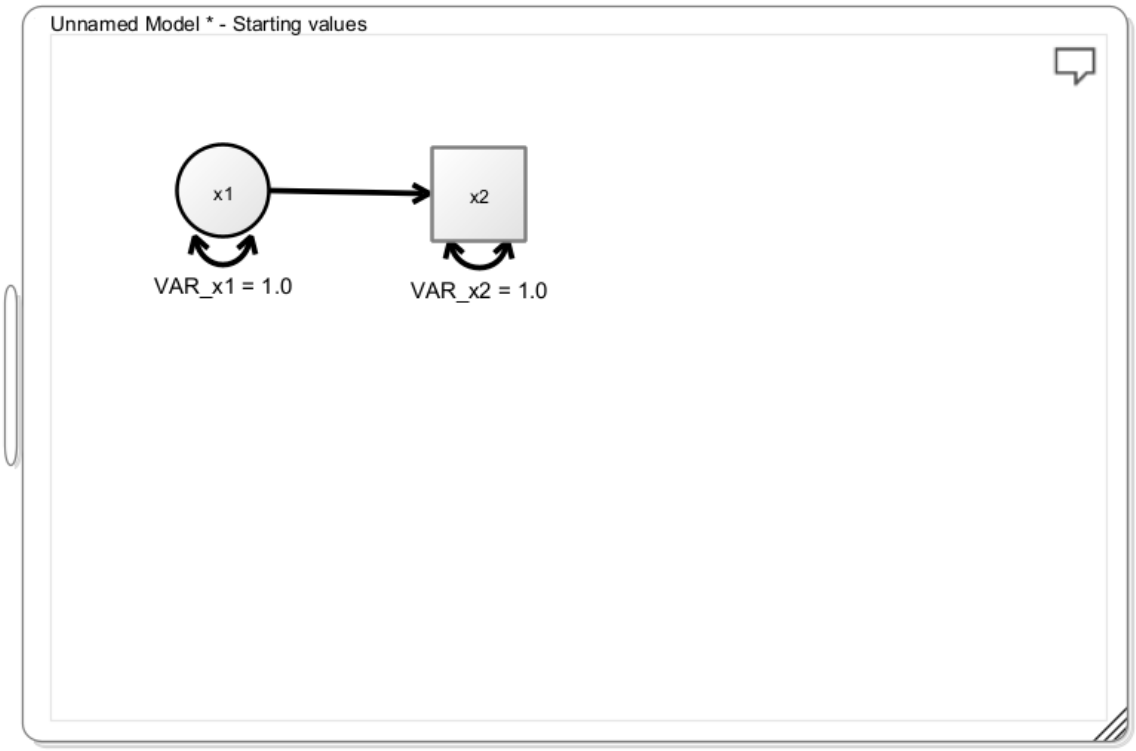
To create a model, either right-click on the empty desktop and select “Create new Model -> Create Empty Model” or double-click on the desktop. This creates an empty model view

You can now add variables, by either right-clicking on the empty view and selecting “Create Variable”. Then you can either add a latent variable (depicted as a circle), an observed variable (depicted as a rectangle), a constant variable (depicted as a triangle). Beyond standard RAM notation, you can also use XRAM notation, in which you can add a multiplication node that allows you to model products of variables (Boker et al. 2023). This allows models including latent interactions. As a short-cut, you can simply double-click on an empty space in the model, to create a latent variable or double-click while holding down the SHIFT-key to add an observed variable.

To add a path between two variables, press the right mouse button on a variable, hold it down while dragging the path on to the target variable, then release the mouse button. This creates a regression path with a value fixed to "1" between the variables. When you hold down SHIFT while doing so, you create a covariance path. Right-click the path and select "Free Parameter" if you want this path to be freely estimated.



Try to add a latent and an observed variable and connect them with a path from the latent to the observed variable, and your model should look like this:

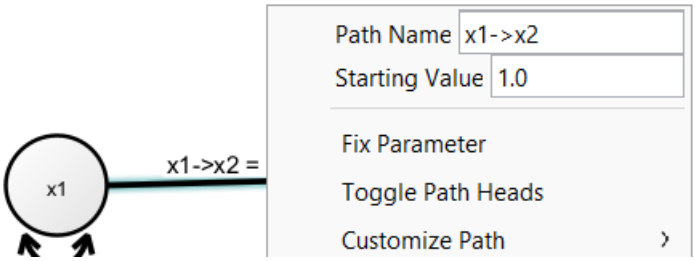


3.1 Changing Visuals

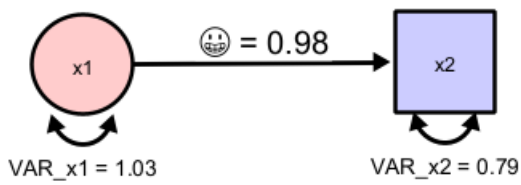
Onyx has a variety of options to adjust the appearance of the graph. Right-click on a node and choose "Customize Variable" to explore various options to adjust font size, line colors, fill colors, etc. The same is possible if you choose "Customize Path" when you right-click on a path.

3.2 Renaming Parameters

If you free a parameter on a path (representing either a variance, covariance, a regression, or a mean), you can change the name of the parameter by right-clicking on the path and then changing the "Path Name". By default, these parameters are named based on the adjacent variables but you can choose your own name here.



Note that Onyx has Unicode support, that is, you can directly use unicode characters such as letters from the greek alphabet. Thus, it would even be possible to use emojis as parameter labels even though I do not recommend that:



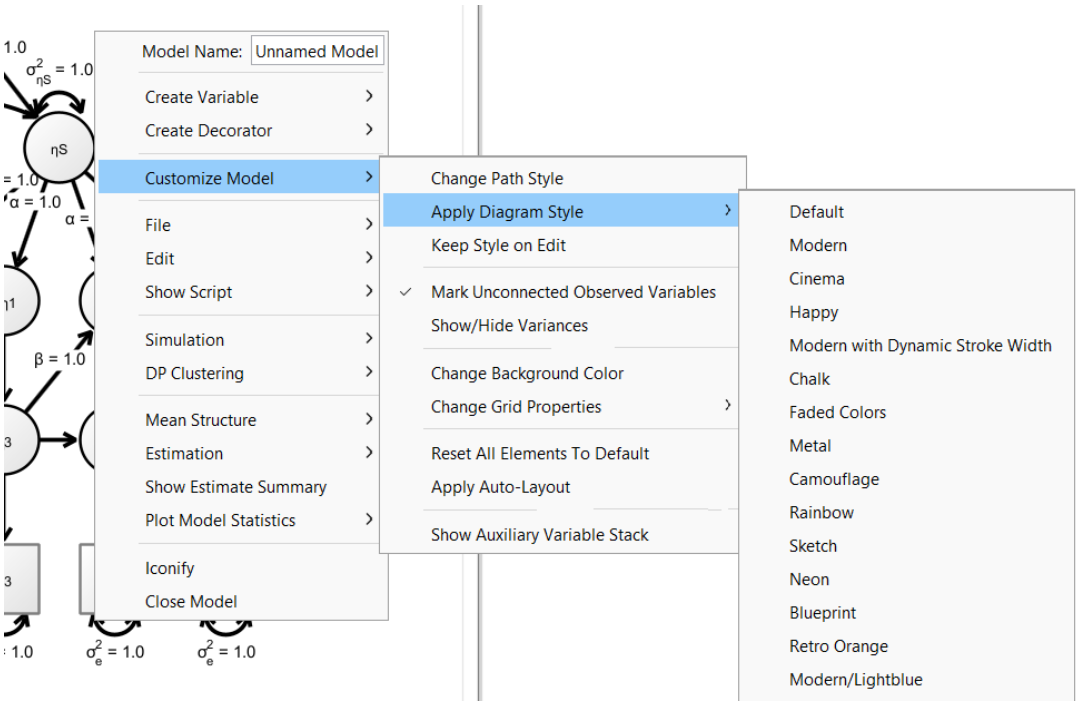
Onyx further supports a very limited subset of LaTeX commands to render parameter labels. This pseudo LaTeX input is triggered by starting labels with a dollar sign. This pseudo input supports lowercase and uppercase greek letters, as well as subscript and superscripts. Here are some examples that are valid parameter names in Onyx:

Input	Output
\\alpha	α
\\beta^2	β^2
\\gamma_{person}	γ_{person}
\\Omega^2_i	Ω^2_i

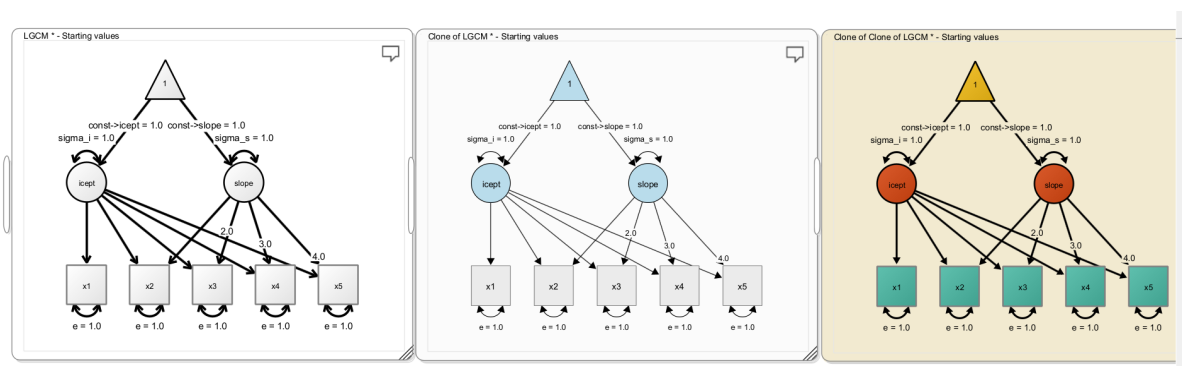
3.3 Diagram Styles

Onyx ships some pre-made styles for your graphs. Choose "Customize Model -> Apply Diagram Style" and then choose from the available choices. Certainly, not all of them are useful for scientific communication but you may want to start with Modern or Modern/Lightblue.

Further, you may want to choose “Keep Style on Edit”, which continues to automatically apply the chosen style to newly added variables or paths in the diagram (otherwise, they are added using the default style).



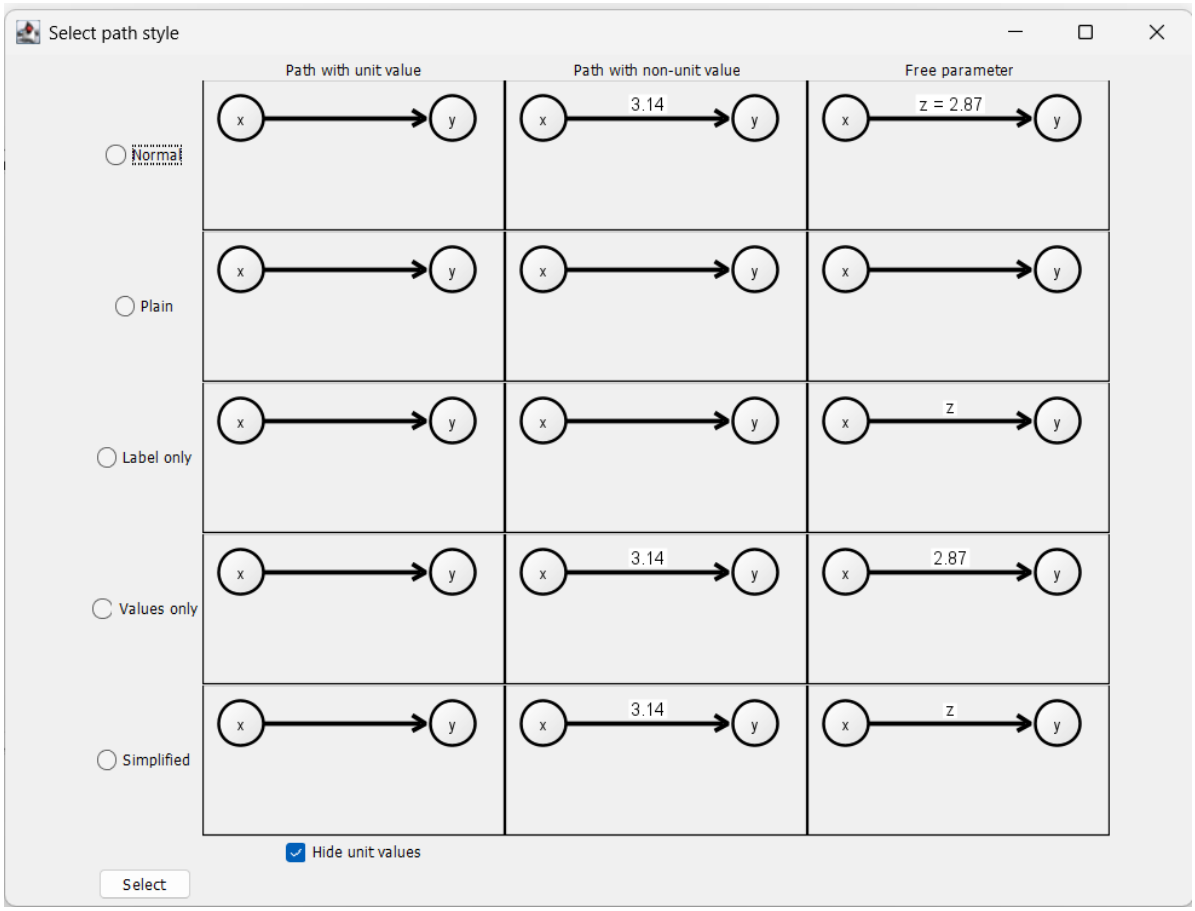
Here are three examples of styles:



You can quickly iterate through the available styles in the actively selected model by pressing CTRL+L (or CMD+L on a Mac).

3.4 Path Styles

By default, Onyx shows path with a value fixed to one without any label. Paths that have a value fixed but different than one are shown with the value as label. Freely estimated parameters are shown with both their label and their value (e.g., "z = 2.87"). You can also choose different path styles by right-clicking on a model, then selecting "Customize Model -> Change Path Style". This opens the following window in which you can select among several alternative options. In particular, the option "Simplified" may be useful to create figures for manuscripts.



3.5 Copy & Paste

Onyx supports copy&paste in diagrams. You can select a set of nodes, right-click the mouse and then select "Edit -> Copy" (or press CTRL+C / CMD+C), which copies all selected nodes and all paths between them to the clipboard. If you select "Edit -> Paste and Rename"

(or press CTRL+V / CMD + V), a copy of these nodes and paths is inserted. Note that all parameter and variable names are automatically renamed to be different from the original variables and paths; otherwise they would be treated as being constrained to identity. If you prefer that choose "Edit -> Paste" to insert exact copies without renaming. If you want to select the entire diagram to copy, press CTRL+A / CMD+A to select all variables before copying and pasting.

3.6 Exercises

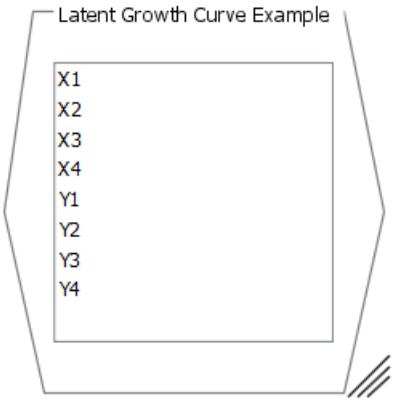
- Create a factor model with one latent factors, four observed variables, residual variances for each factor, a latent factor variance, and factor loadings; fix the first factor loading at "1" and let the others be freely estimated; also freely estimate all residual variances; rename the observed variables to "item1", "item2", …; rename the factor to "factor", rename the factor loadings to lambda1, lambda2, … but use greek letters.
- Apply diagram styles (CTRL+L / CMD + L; press multiple times) and then make your own modifications
- Export the diagram as image file
- Copy the entire factor model and paste it below. Select the original model and flip it vertically (using the short-cut CTRL+F / CMD+F), such that the latent factors are facing each other; then draw a covariance between the two latent factors and give it a blue color

4 Loading Data and Parameter Estimation

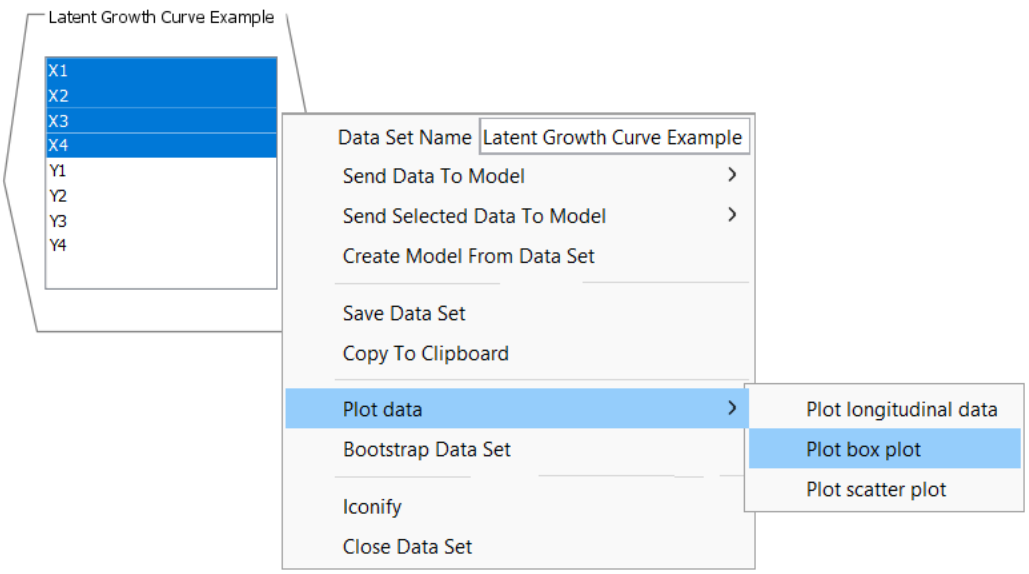
4.1 Loading Data

Onyx supports loading of data from three types of data files, either delimted text files (most typically, the comma-separated value file or CSV), SPSS files, or text files that describe an observed covariance matrix. We recommend using CSV files as the most interoperable exchange format. To load a datafile, you can either right-click on the empty desktop and select "Load Model or Data". A file dialog will appear from which you can either choose a CSV file, an SPSS data file, or a text file with a covariance matrix. We also provide some toy datasets, which you can load from the same context menu by clicking "Load tutorial data" and then by selecting one of the four shipped datasets.

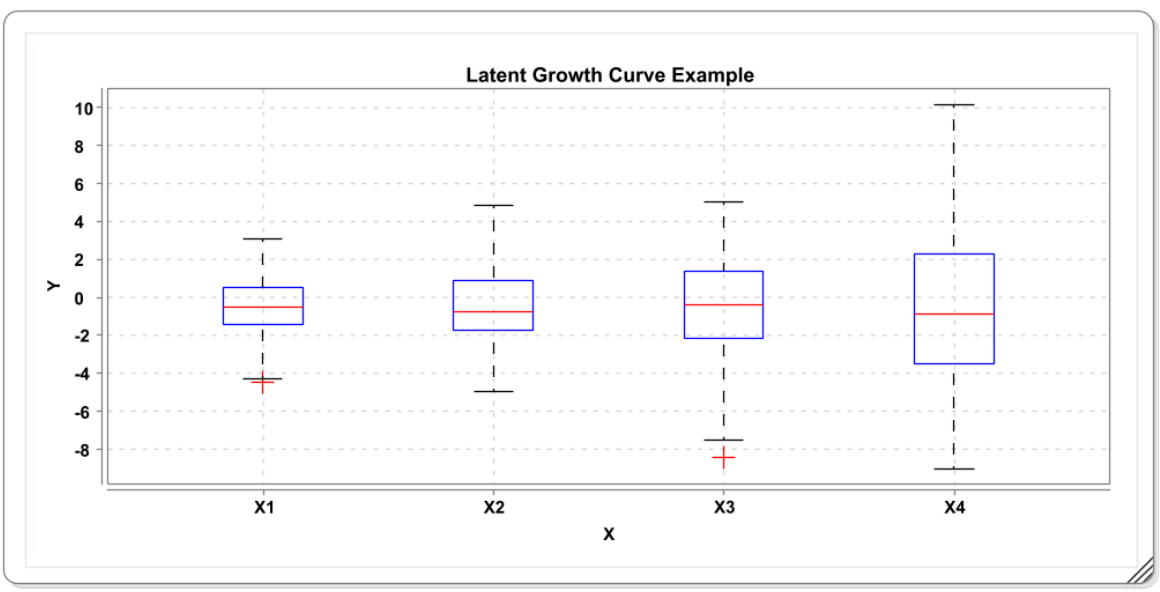
Datasets appear as hexagonal shaped data views:



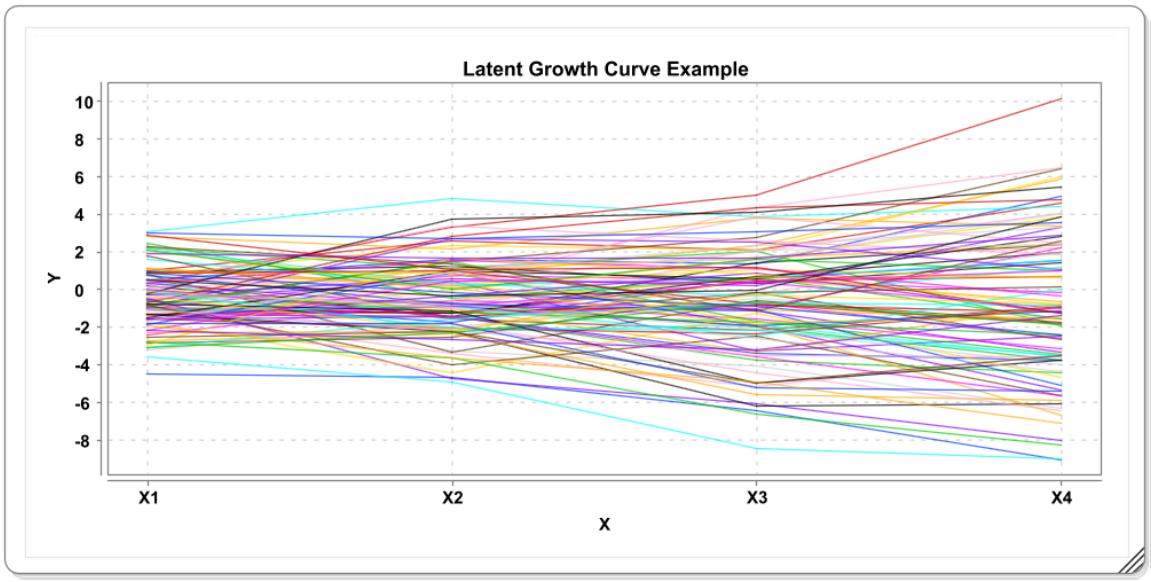
The data view contains a list of all variables in the dataset. You can use some basic plotting functionality to look at these data. For example:



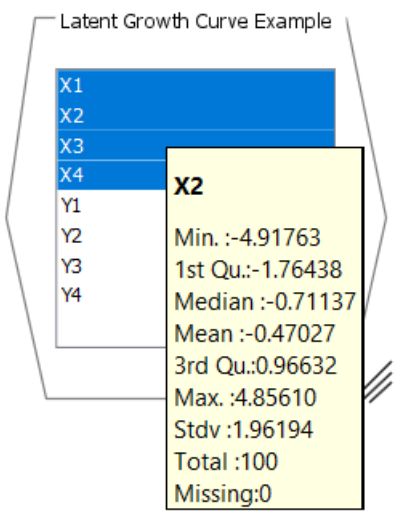
This will create a plot like this:



Also try out the other two plotting options. For example, the longitudinal plot:

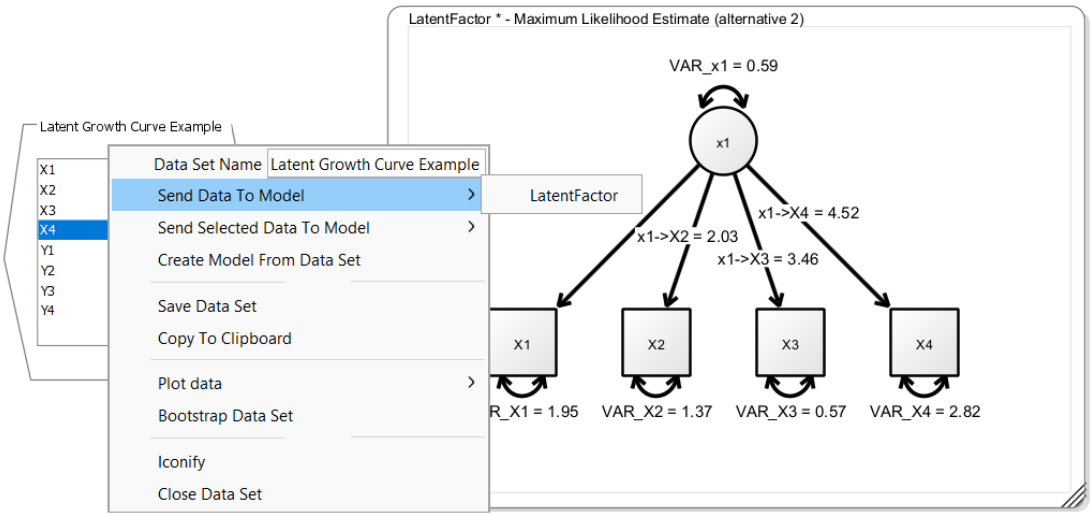


You can also hover the mouse over a variable in the data view and a popup window will appear that contains basic statistics. It's a good way to check these statistics to see whether Onyx loaded the data correctly:



4.2 Connecting Data and Models

To start model estimation, you need to connect all variables in a model with variables from a dataset. To do so, you can right-click on the data view, select "Send Data To Model" and then select the name of the model you want to connect the variables to. Note that Onyx depicts variables that are not yet connected with a light gray border color and once they are connected they get their regular border color. Once all variables in a model are connected to a variable, estimate in Onyx is started and once the estimation process converged, the resulting parameter estimates are directly shown in the Onyx model.



4.3 Model Creation Shortcuts

The fastest way to start modeling is to simply drag and drop one or more selected variables from the data view to a model view. This will create new observed variables with matching names that are already connected to the data set.

4.4 Exercise

1. Load the tutorial dataset "Confirmatory Factor Example"
2. Plot the data using boxplots and a scatterplot of x1 and x2

3. Create an empty model, create variables with names matching those of the dataset (e.g., simply drag and drop the variables from the data view to the model view)
4. Specify a common factor and see how Onyx refits the model as you modify the graph

5 Model Fitting

5.1 Model Fit

In SEM, the fit function to obtain maximum likelihood estimates under multivariate normality of the observations and assuming no missing data is:

$$F_{\mathrm{ML}} = \log|\Sigma(\theta)| + \mathrm{tr}\left(\mathbf{S}\,\Sigma(\theta)^{-1}\right) - \log|\mathbf{S}| - p$$

where Σ is the model-implied covariance matrix and S is the observed covariance matrix. This can be expanded to also include means.

Assuming H_0 that there is no misfit is true, the following test statistic has an asymptotic χ^2 distribution

$$\chi^2 = (N-1)\,F_{\mathrm{ML}}$$

5.2 Model Fit

- χ^2 is the fundamental model fit measure as discrepancy between model-implied (co)variances and means and observed (co)variances and means
- Significance test of close fit is usually not recommended as it has very high power; instead we usually favor effect-size-type metrics
- CFI: discrepancy between an independence model and the hypothesized model; *CFI asks: How much better does my model fit compared to a model that assumes no relationships at all?*
- RMSEA: estimates how much misfit per degree of freedom your model has in the population. *RMSEA asks: "Is the model achieving good fit without overusing parameters?*

5.3 Residuals

Residuals represent the difference between what the model predicts and what is actually observed in the data. More specifically, SEM works by reproducing the observed covariance (or correlation) matrix using a model-implied covariance matrix. Residuals are simply the element-wise differences between these two matrices. If the model fits well, these residuals should be small—ideally close to zero—because the model is successfully capturing the relationships among variables.

Residuals are useful because they provide localized information about model misfit. Global fit indices like chi-square, CFI, or RMSEA tell you whether the model fits overall, but they don't tell you where it fails. Residuals, in contrast, point to specific pairs of variables where the model does a poor job reproducing the observed relationship. For example, if two indicators have a large positive residual, it means their observed correlation is higher than what the model predicts, suggesting that something is missing, such as a residual correlation or an additional factor.

5.4 Raw and Standarized Residuals

There are different types of residuals in SEM. Raw residuals are simple differences between observed and implied covariances, while standardized residuals scale these differences by their expected variability, making them easier to interpret across different variable pairs. Standardized residuals are often preferred because they help identify which discrepancies are unusually large relative to sampling variability.

Importantly, residuals are not just diagnostic tools – they also guide model refinement. Large residuals can suggest theoretically meaningful modifications, such as allowing two indicators to correlate due to shared wording or content (e.g., shared method variance). However, they must be interpreted carefully: adding parameters purely to eliminate residuals can lead to overfitting and models that do not generalize well.

5.5 Model-implied and observed covariances

A summary measure of the misfit in the residuals is the standardized root mean square residuals, which is:

$$\text{SRMR} = \sqrt{\frac{2}{p(p+1)} \sum_{i \leq j} \left(r_{ij} - \hat{r}_{ij}\right)^2}$$

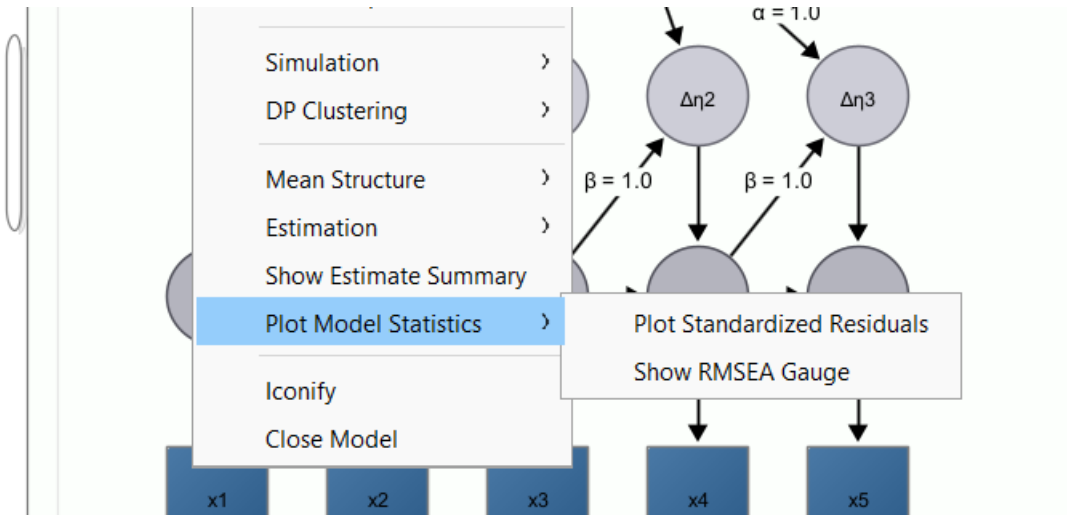
where

r_{ij} are the observed correlations $\hat{r_{ij}}$ are the model-implied correlations

The exact computation differs across programs. Some programs standardize such that the means are included and some don't. Some programs standardize such that the diagonal elements are included and some don't. Note that Onyx standardizes the observed covariance matrix and then the model-implied covariance matrix each with their own variances. Thus, Onyx SRMR does not include misfit in the means or the residual variances – just the covariances.

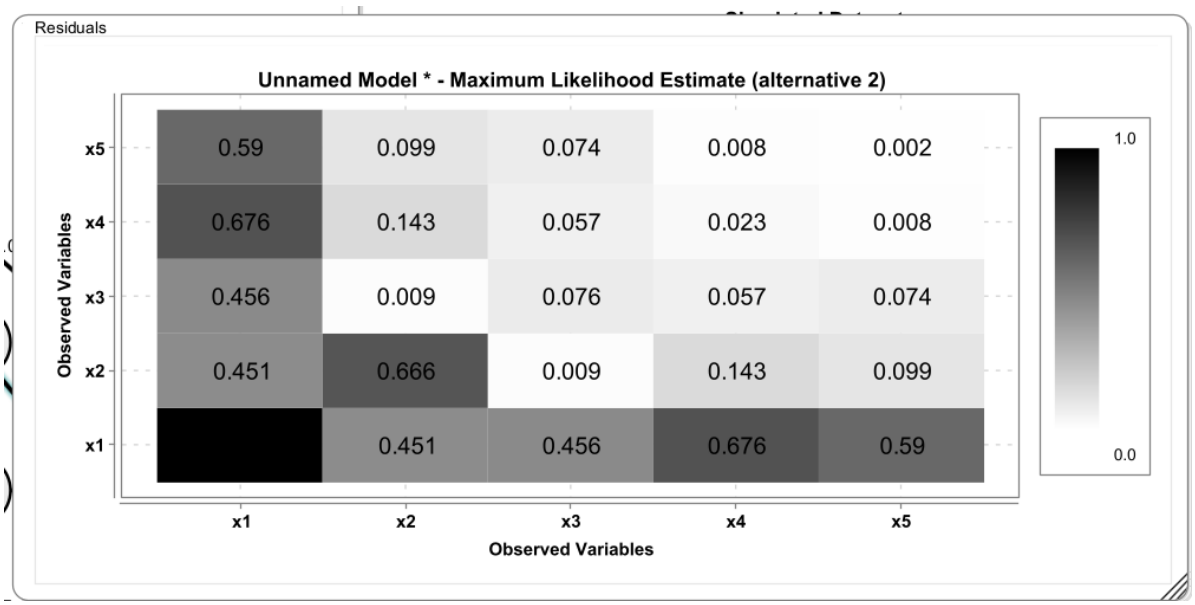
5.6 Model Fit Plots

There are two new plotting features to obtain visual information about model fit. Right-click on a model and select "Plot model statistics". Then choose "Plot Standardized Residuals"



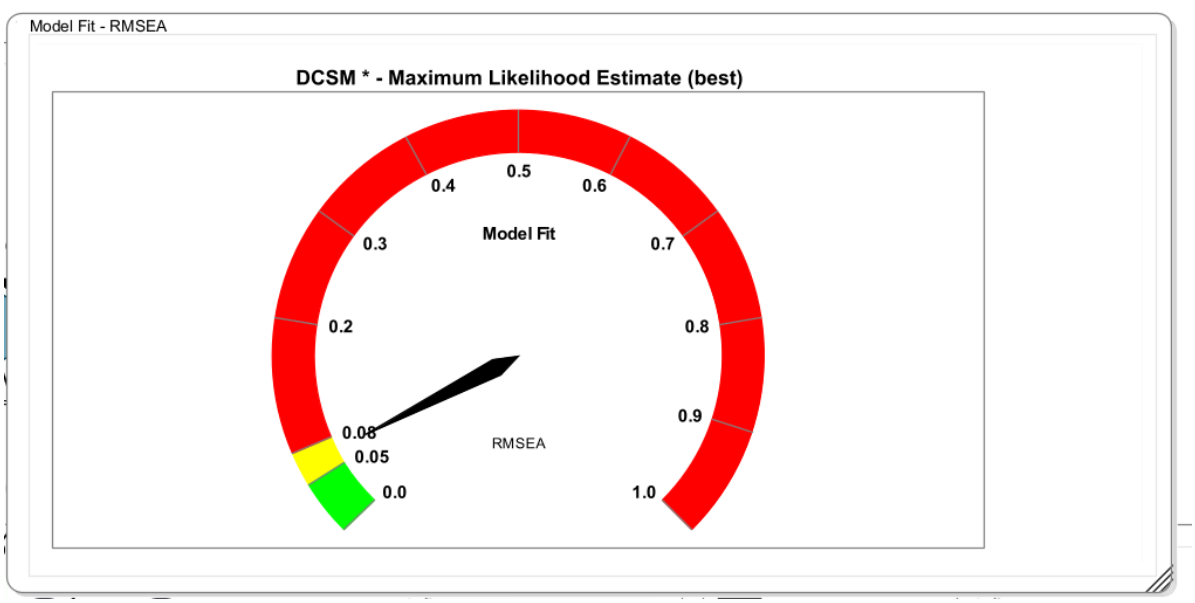
5.7 Residual Plots

This opens the standardized residuals plot. This plot show standardized residuals that are slightly different than what is used to compute the SRMR. These standardized residuals are computed such that they also represent misfit in the diagonal of the matrix, that is, the residual variances. This is achieved by standardizing both model-implied covariances and observed covariances with the variances of the observed covariance matrix.



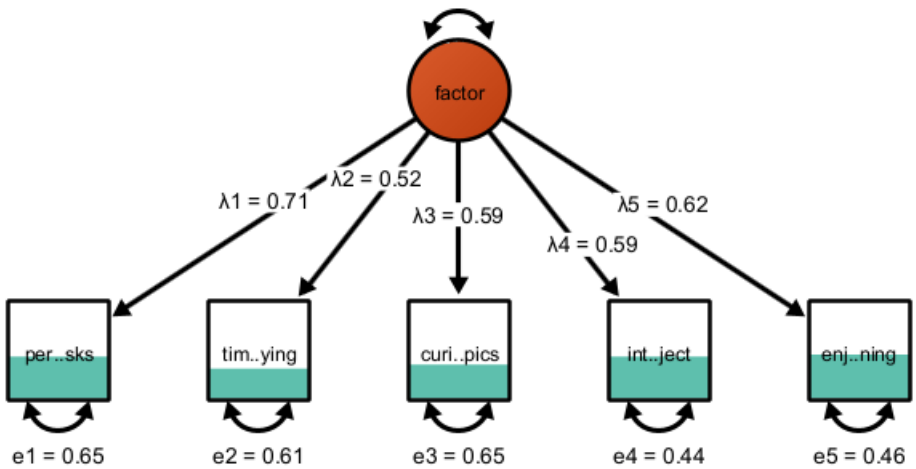
5.8 Model Fit Gauge

A further visualization of model misfit is the RMSEA gauge. This is a gauge that shows the RMSEA and shades some region of the RMSEA according to whether those ranges were proposed as ranges of acceptable or good fit.



5.9 Visualizing Explained Variance

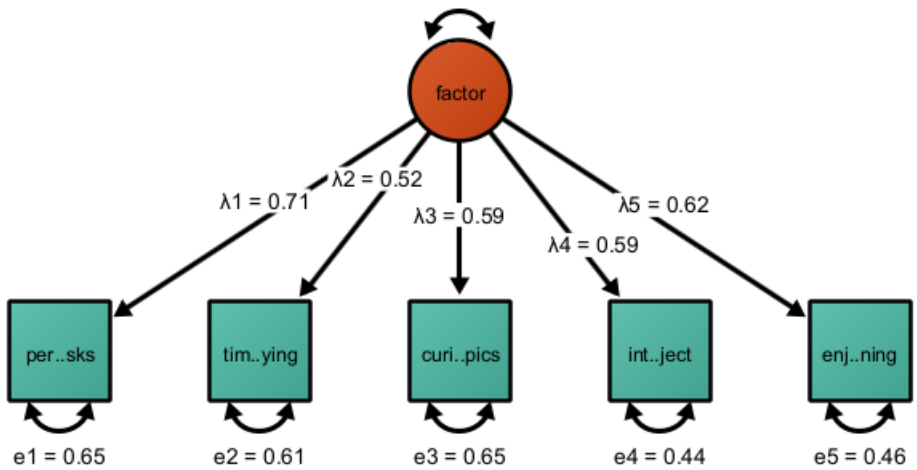
For observed variables, you can select a fill-style that reflects the percentage of explained variance in a variable. Right-click on a variable or a selection of variables, click "Customize Variable", then "Select Fill Style", then "Bokers R2". This applies a fill-style that fills the selected variable only up to the level that corresponds to the variance explained by other variables in the model. The unexplained part is not filled. This gives a good quick overview, for example in factor models, how well an item works. Here is an example in a factor model with five indicators:



5.10 Exercise

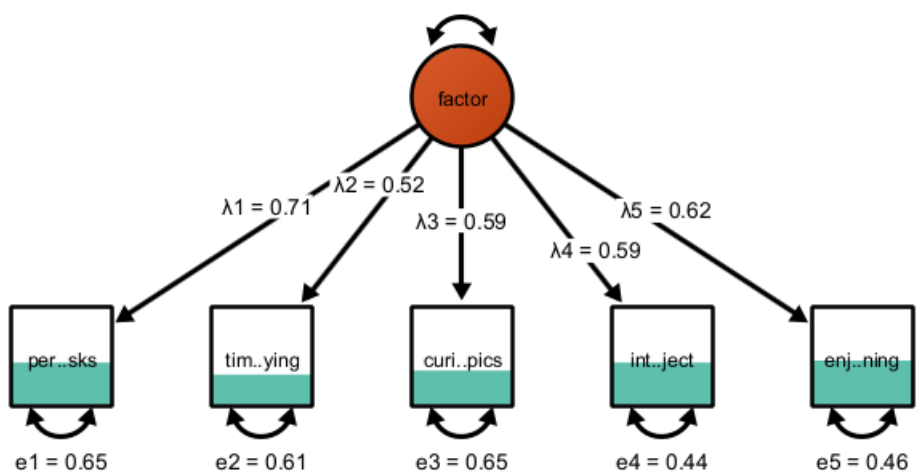
- Open the dataset "academicmotivation.csv" in Onyx
- This is a simulated dataset of a common factor of academic motivation that is measured with five indicators: persistence_on_tasks, time_spent_studying, curiosity_about_topics, interest_in_subject, and enjoyment_of_learning
- Fit a common factor model (just covariances; no means); apply Boker's R2 style to the observed variables
- Inspect model fit, model-implied covariances, and residuals (using the plot)
- Given these statistics and the information that interest_in_subject and and enjoyment_of_learningtion tap more into intrinsic_motivation than the others, how could we modify the model?

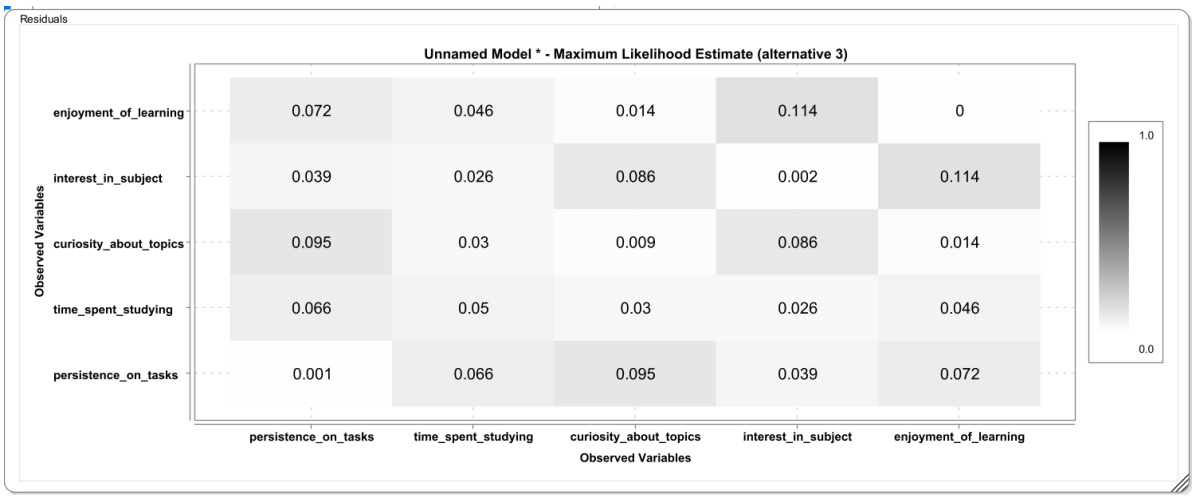
5.11 Solution



Model fit is $\chi^2 = 21.033$, RMSEA=0.106, CFI=0.106, SRMR=0.055. RMSEA and CFI are outside regions that are usually deemed acceptable.

Applying Boker's R2 style, this yiels:





The large discrepancy in the correlation of enjoyment_of_learning and interest_in_subject may be an indication that these items also capture something different than the other items. In light of a theory (intrinsic vs academic motivation), this can be used to modify the factor model to include a residual correlation (which is akin to a second common source of variance) to the model. By adding another free parameter, model misfit in the present sample is reduced. Note that such model modifications after having seen the data should always be reported as exploratory.

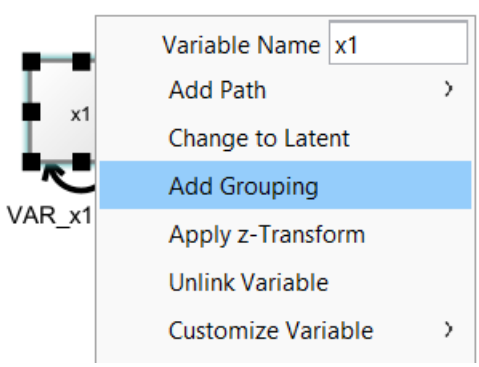
6 Multi-Group Models

Multi-group models in Structural Equation Modeling (SEM) are an extension of standard SEM techniques that allow researchers to test whether relationships between variables are consistent or different across two or more groups. In the context of Psychology, these groups might represent categories such as gender, age cohorts, cultural backgrounds, or clinical vs. non-clinical populations. By estimating the same model simultaneously in multiple groups, researchers can formally assess measurement invariance (whether constructs are interpreted similarly across groups) and structural invariance (whether the relationships between constructs differ). This is especially valuable because it ensures that observed group differences reflect true psychological distinctions rather than artifacts of measurement bias. Overall, multi-group SEM provides a rigorous framework for comparing theoretical models across populations, strengthening the validity and generalizability of psychological findings.

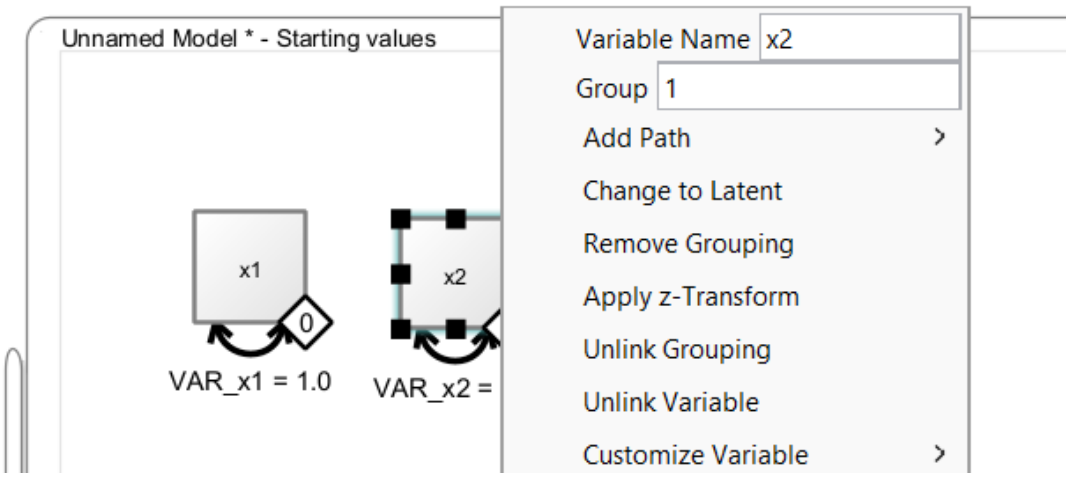
6.1 Multi-group models in Onyx

Onyx supports multi-group models, in which we use a grouping variable that encodes group membership between individuals. Onyx has its own approach to multigroup modeling which works with filters on observed variables that filter out subsets of observations that should be modeled by a given observed variable.

By right-clicking on an observed variable, you can select "Add Grouping" to add a filter to one or more selected observed variables.



Then, a little diamond grouping indicator appears on that variable. The value inside the diamond reflects the group membership. To change group membership, right-click the diamond and change the "Group" value:



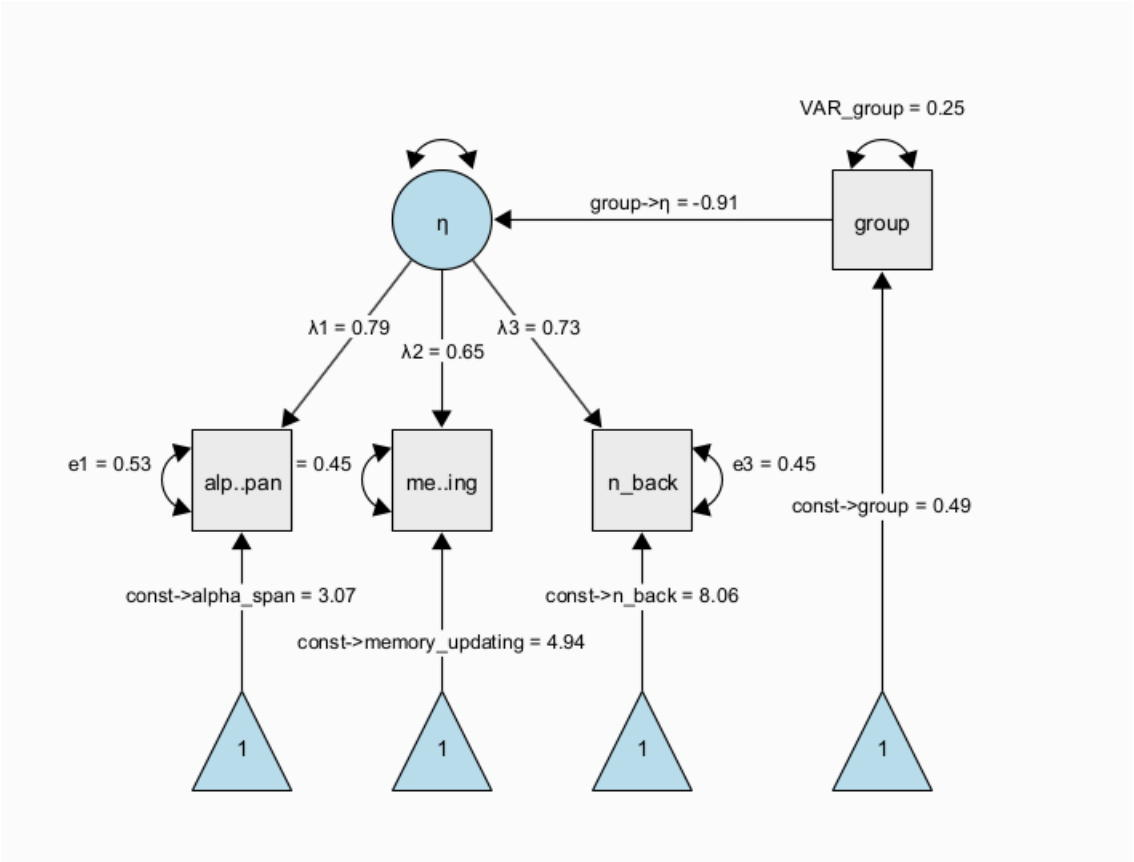
6.2 Latent Mean Differences

The diamonds are variable containers, that is, you can drag a discrete-valued grouping variable from a dataset on to the diamond. As long as the diamond is not filled (white fill color), it is not connected. Once it is connected, it becomes filled (black color), that is, it is now associated with a grouping variable in the dataset.

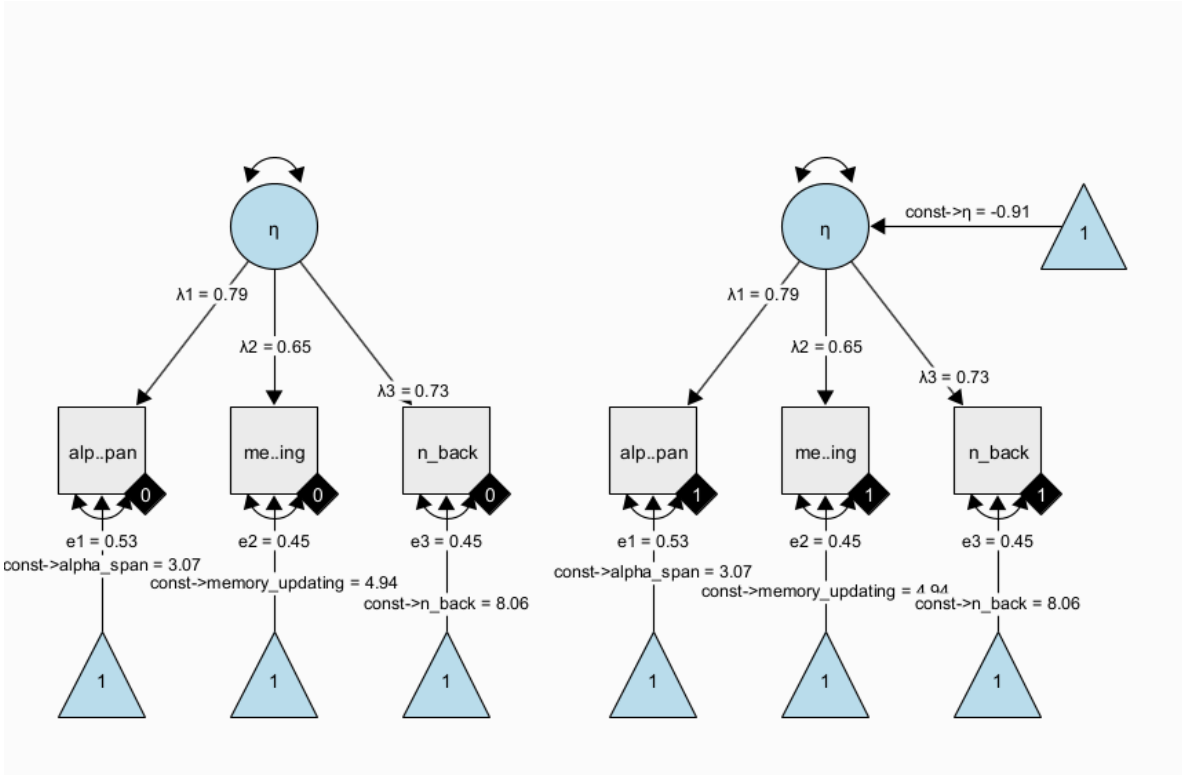
7 Applied Example: Latent Means

Models of latent mean differences are like the classical t-test or ANOVA but estimated on latent constructs.

Figure 1 shows a path diagram representing one possibility to set up a latent mean difference model when there are multiple, noisy measurements of the same underlying latent construct. This model is a MIMIC (multiple-indicators multiple-causes, Hancock 2001) model, in which we estimate a measurement model of a latent trait η across two groups that are encoded via a dummy group indicator. The latent trait is measured through multiple observed indicators (here, $x1$ to $x3$) with indicator-specific residual variance terms, which allows to separate error variance from true-score variance. If correctly specified, this approach allows us to obtain more precise estimates of the latent trait purged of measurement error and unreliability (DeShon 1998). In this specific instance, the regression path from the dummy-coded group indicator to the latent variable directly represents a standardized mean difference because the scale of the latent variable was fixed to have a variance of one. Note that the MIMIC model assumes that measurement invariance holds, that is, the factor loadings, item intercepts, and item residual variances are assumed to be the same across groups.



A more flexible model specification is also possible; using a multiple-group factor model, we can estimate the same factor model in each group separately and empirically test whether measurement invariance holds; then, we can continue to test whether there is a difference in latent factor means across groups. This is also known as the SMM [structured means modeling; Hancock (2001)] approach:



Note that all parameters with the same name have identity constraints; in particular, this means that parameters with the same name are identical across groups. In the path diagram shown above, all parameters are identical across groups. This implements a very strong assumption of measurement invariance across groups and identity of magnitude of variance (= individual differences at the latent level) across groups. Again, this restricted version of the SMM model is identical to the above dummy-coded MIMIC approach. However, the SMM allows to also freely estimate parameters between groups and perform tests of whether some of the restrictions are supported by the data or not.

7.1 Exercises

- Load the dataset "factor_model_memory.csv"
- Use the "Latent Means Model" wizard to create both the SMM and the MIMIC approach model to estimate models of latent mean differences
- Freely estimate the latent variance in the second group of the SMM approach
- Copy the SMM model and freely estimate the factor loadings across groups; run a likelihood ratio test between the models as a test of measurement invariance

7.2 Solution

TODO

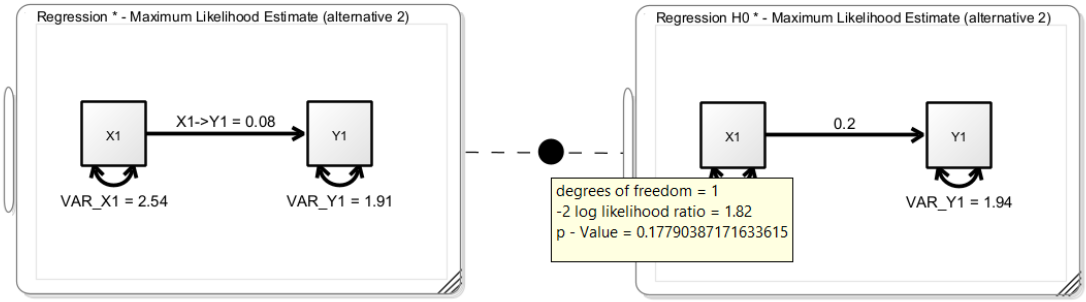
8 Model Comparison

8.1 Model Comparison

Model comparison in Structural Equation Modeling is a central technique for evaluating how well different theoretical specifications fit the observed data. In Psychology, researchers often compare **nested models**, where one model is a constrained version of another (e.g., fixing certain parameters to zero or imposing equality constraints). This nesting allows for the use of **likelihood ratio tests (LRTs)**, which formally assess whether the more complex model provides a significantly better fit than the simpler one. By examining changes in model fit (typically via differences in chi-square statistics for nested models) researchers can determine whether added parameters meaningfully improve explanatory power or whether a more parsimonious model is sufficient. This approach is particularly useful for testing theoretical assumptions, such as whether specific pathways are necessary or whether constraints (e.g., equal effects across groups or time points, measurement invariance, additional predictors) are justified, thereby supporting more precise and theory-driven conclusions.

8.2 Model Comparison using Chi-Square-Tests

Onyx supports model comparison for nested models based on the chi-square test. To compare two models, you can add a model comparison of two models that is represented as an edge between two models (similar to a path between two variables). The model comparison is created analogously to creating a path between two variables. Press the right mouse button on an empty space of one model, the mouse pointer onto the second model, and release the mouse button. A model comparison edge will appear between the two models – hovering the mouse over the circle will show a popup window with the results of the model comparison including the test statistics (-2 log likelihood ratio) and a p value:



In this example, we see a simple regression model (without means) on the left-hand side. The unstandarized regression coefficient is 0.08. The right-hand model is a clone of the left model with the regression coefficient fixed to 0.2; this corresponds to a null hypothesis that the true regression coefficient is 0.2. The p-value of the model comparison is not significant, thus we cannot reject the restricted null model. We did not find evidence that the true regression coefficient is different from 0.2.

8.3 Comparing Non-Nested Models

Onyx performs a nesting test. If models are not nested, the likelihood ratio test is not applicable. In these instances, there are various heuristic approaches that are usually taken. By default, Onyx shows the difference in AIC values and shows which model is favored according to AIC. Other researchers prefer to compare CFI and RMSEA and choose the model which has better fit according to these criteria.

8.4 Holzinger Swineford

From the description provided in the lavaan package:

> The classic Holzinger and Swineford (1939) dataset consists of mental ability test scores of seventh- and eighth-grade children from two different schools (Pasteur and Grant-White). In the original dataset (available in the `MBESS` package), there are scores for 26 tests. However, a smaller subset with 9 variables is more widely used in the literature (for example in Joreskog's 1969 paper, which also uses the 145 subjects from the Grant-White school only).

The commonly used factor model for these 9 variables consists of three latent variables each measured by three indicators:

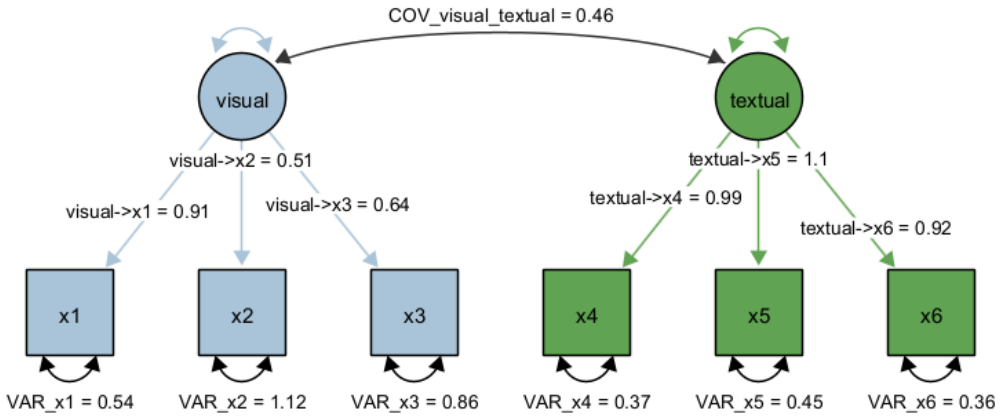
- a *visual* factor measured by three variables: `x1`, `x2` and `x3`
- a *textual* factor measured by three variables: `x4`, `x5` and `x6`
- a *speed* factor measured by three variables: `x7`, `x8` and `x9`

8.5 Exercise

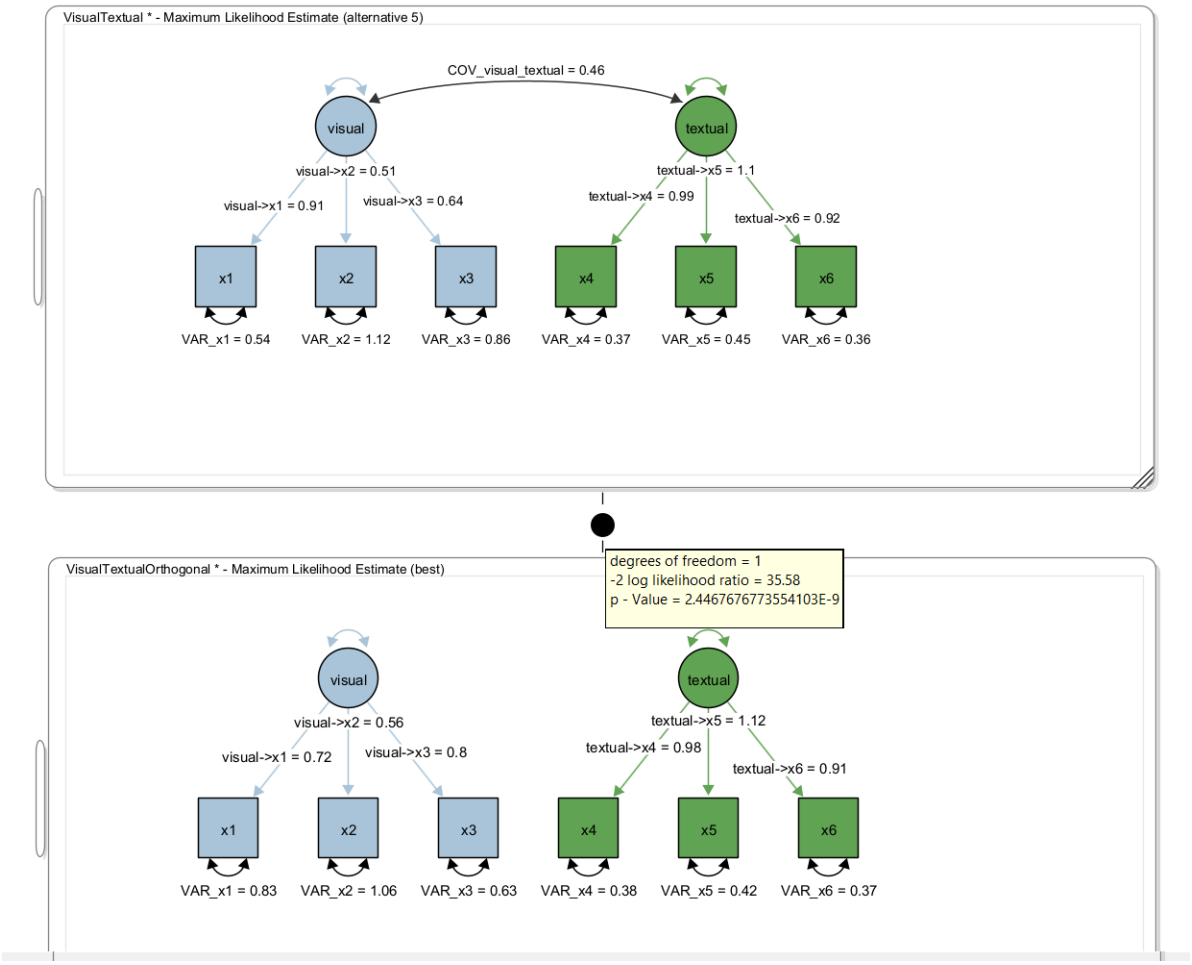
1. Load the dataset `HolzingerSwineford1939.csv`
2. Create a latent factor model with the first six observed variables (x1 to x6); allow a correlation of the two latent factors; name the model "VisualTextual"
3. Make a copy of the model ("edit -> clone model") and rename it to "VisualTextualIndependent"
4. Modify the model "VisualTextualIndependent" by removing the correlation between the two latent factors
5. Use a likelihood ratio test to compare the two models. Which model is favored? Is there evidence for a correlation of the textual and visual factor?
6. Obtain latent variable factor scores and plot them in a scatterplot

8.6 Solution

First, we load the dataset, we select variables x1 to x6 (e.g., by selecting x1, holding down the SHIFT key and then selecting x6 in the variable list of the dataset view). Next, we create two latent factors and draw the factor loadings. Last, we fix the latent variances to one and estimate all factor loadings freely; alternatively, we could fix one factor loading to one and estimate the latent variances freely. However, by fixing the latent variances to one, we can directly interpret the latent covariance as correlation. The path diagram should look like this:



Next, we clone the model (using Edit -> Clone Model) and rename the cloned model. Then we right-click the covariance in the cloned model and select "Delete Path". Then, create a model comparisong edge between the two models. Hover the mouse over the filled circle of the model comparison edge and the result of the model comparison appears:



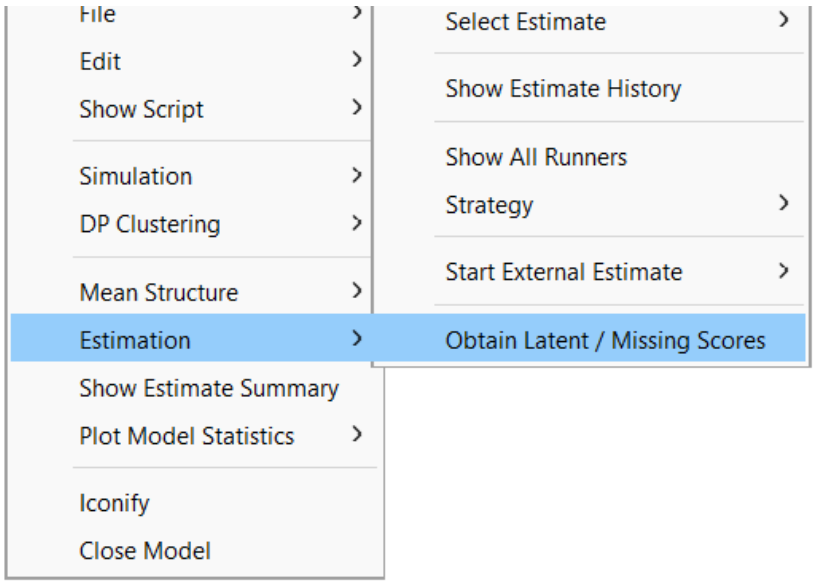
Here, we obtained a test statistic of 35.58 for the likelihood ratio test with one degree-of-freedom (because a single parameter was removed, ie., set to zero). The p value is <0.001 and thus, the model with the extra parameter (ie., the covariance) significantly fits better than the model without the covariance.

9 Factor Scores

Factor score estimation is a post-estimation step in SEM that involves deriving individual-level scores on latent variables from the fitted model. Ideally, any kind of inference should be made using the SEM instead on estimated individual-level scores because the estimation introduces estimation error. For example, if one is interested in the correlation of two latent factors, this correlation should be modeled and model-based estimates should be used for the inference. Common approaches include regression (or "Thurstone") scoring, Bartlett scoring, and empirical Bayes methods, each trading-off bias and efficiency. Importantly, factor scores are not uniquely determined—they depend on the chosen estimation method and model specification—so they should be interpreted as approximations rather than exact measurements of the latent constructs. In applied work, factor scores are often used for secondary analyses, such as predicting outcomes or classifying individuals, but their uncertainty and potential indeterminacy should always be taken into account.

9.1 Factor-score estimation in Onyx

In Onyx, you can right-click on a fitted model view and choose "Estimation -> Obtain Latent / Missing Scores". This creates a new dataset, which imputes all missing variables using a maximum-likelihood-based approach; this includes all latent variables, which are treated as completely missing (as they were not observed):



This new dataset can either be saved or be used to create plots.

9.2 Exercise

For the exercise, we will use the Holzinger Swineford dataset.

9.2.1 Holzinger Swineford

From the description provided in the lavaan package:

> The classic Holzinger and Swineford (1939) dataset consists of mental ability test scores of seventh- and eighth-grade children from two different schools (Pasteur and Grant-White). In the original dataset (available in the MBESS package), there are scores for 26 tests. However, a smaller subset with 9 variables is more widely used in the literature (for example in Joreskog's 1969 paper, which also uses the 145 subjects from the Grant-White school only).

The commonly used factor model for these 9 variables consists of three latent variables each measured by three indicators:

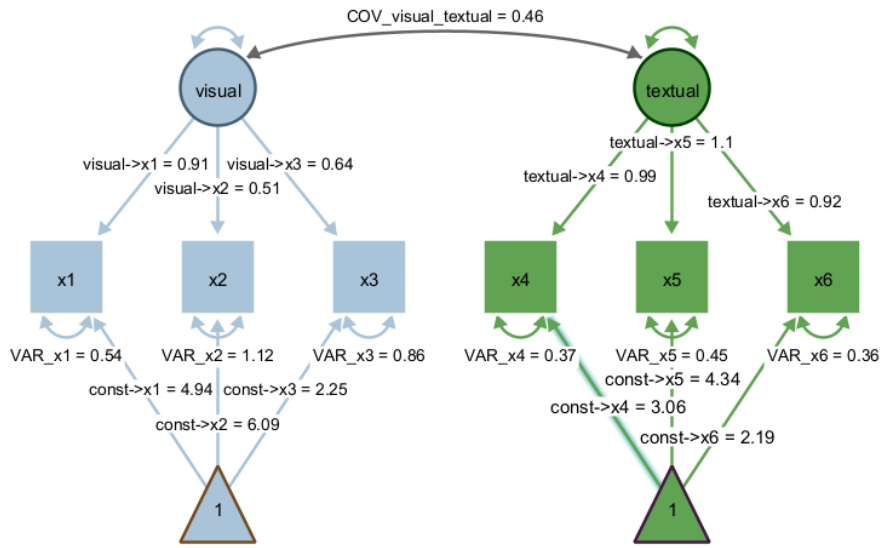
- a visual factor measured by three variables: x1, x2 and x3
- a textual factor measured by three variables: x4, x5 and x6
- a speed factor measured by three variables: x7, x8 and x9

9.2.2 Tasks

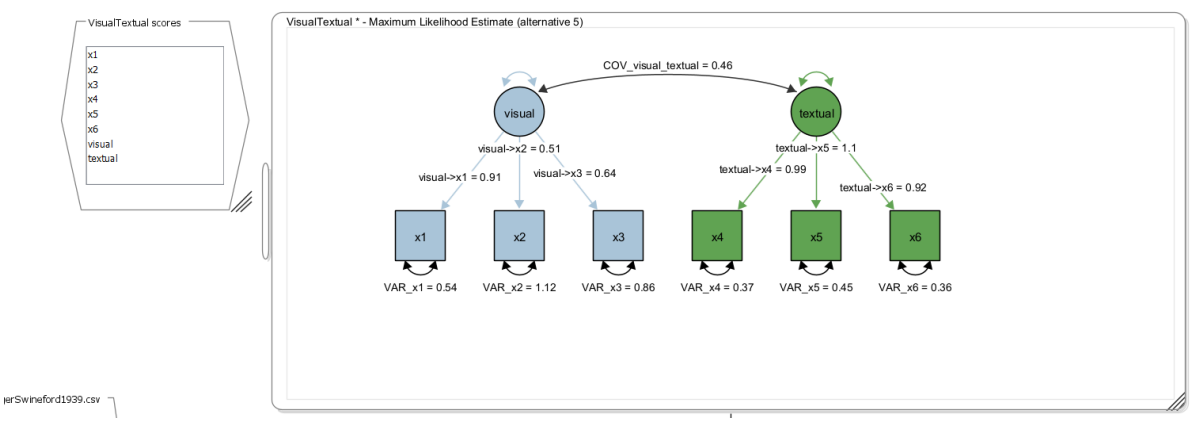
1. Load the dataset `HolzingerSwineford1939.csv`
2. Create a latent factor model with the first six observed variables (x1 to x6); allow a correlation of the two latent factors; name the model "VisualTextual"
3. Obtain factor scores of the two latent variables
4. Plot a scatter plot of the estimated factor scores from the visual and textual factor. What do you see? How does this relate to the latent correlation between the two factors?

10 Solution

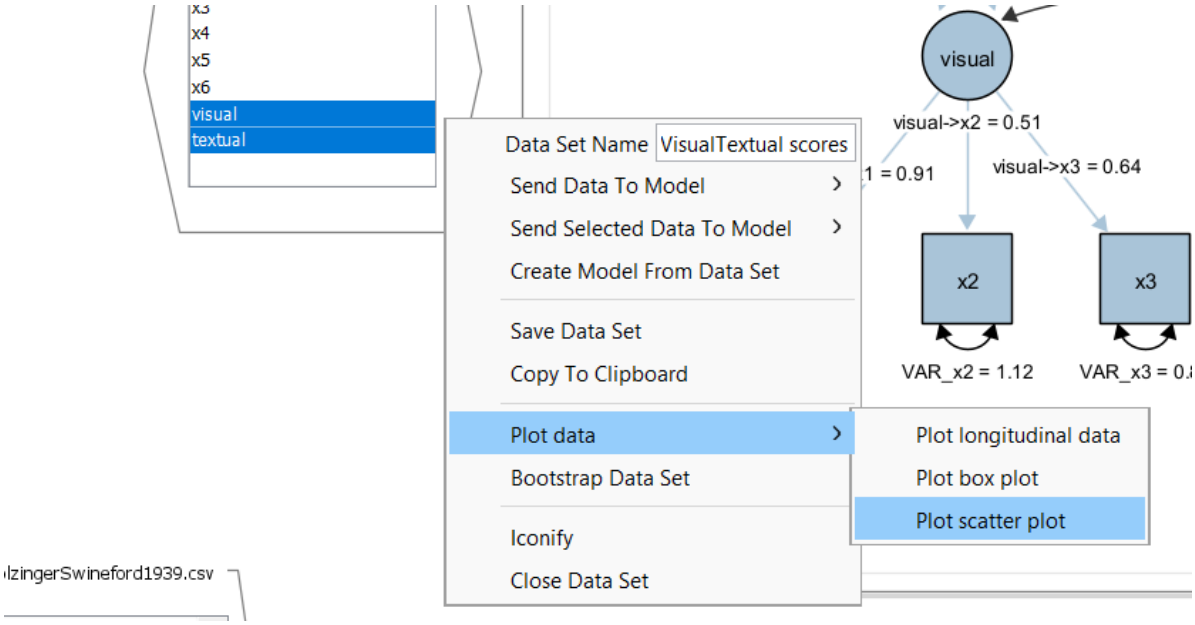
First, create the latent factor model. It should look like this:



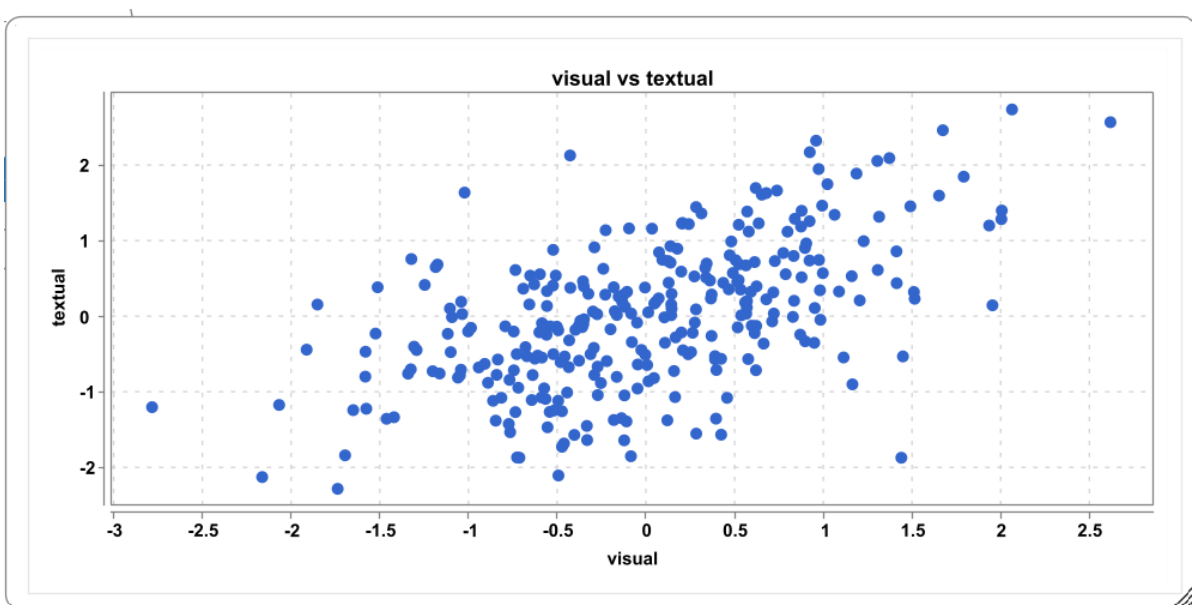
Next, obtain the factor score estimates – this opens a new dataset including the estimated scores as new variables; the names correspond to the names of the latent factors:



Then right-click on an empty space in the dataset view, select plot and then scatterplot:



This creates a scatterplot:

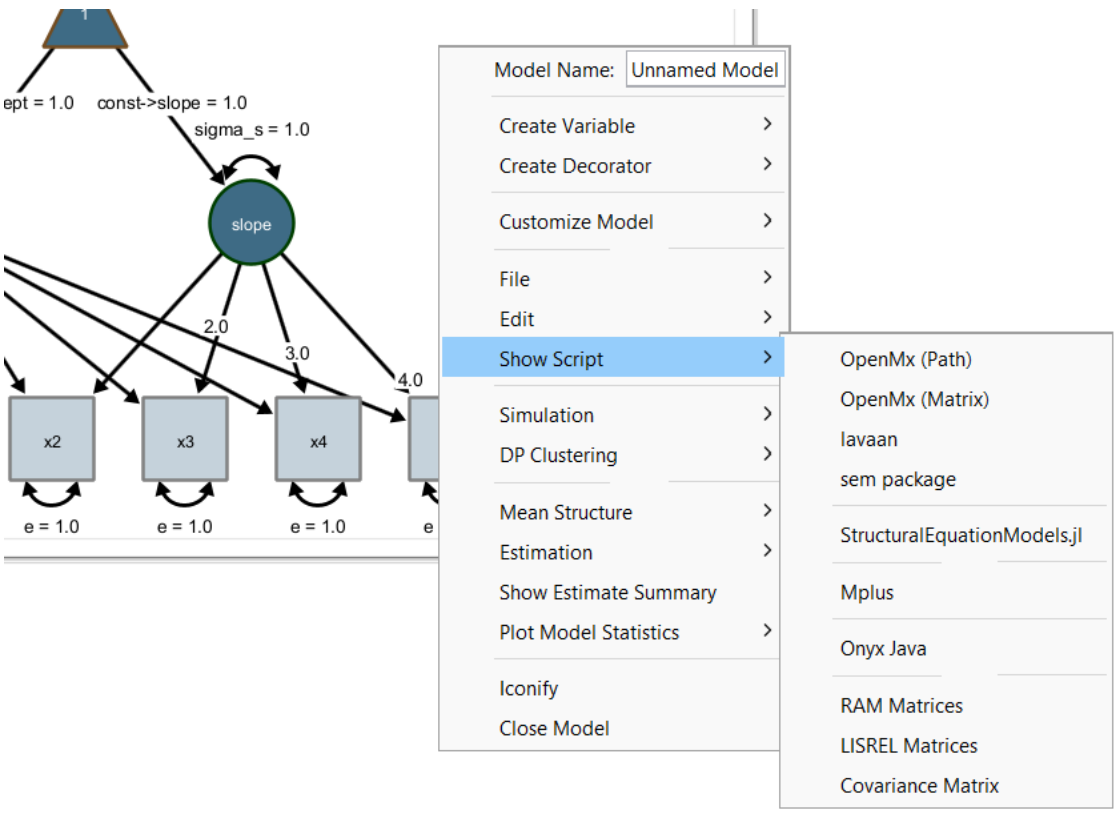


Here, we can see a positive correlation of medium size. This corresponds to the latent factor score correlation in the model. Note, however, that the model-based correlation is a better estimate (less biased and more precise) than estimating a correlation on the estimated factor scores.

11 Exporting Model Syntax

11.1 Syntax Export

Onyx supports exports of models to syntax specifications of other languages. This way, Onyx can be used for graphical model specification and other programs can then be used if their more advanced features are needed for estimating models (e.g., other estimators or non-normally distributed variables, etc.). To view model syntax, right-click a model and choose "Show Script" and select a syntax of your choice. Importantly, this includes the widely used R packages OpenMx and lavaan but also commercial software such as Mplus:



Importantly, the model syntax view remains linked to the model, that is, if the model graph is changed (e.g., by adding or removing variables or paths in the path diagram), the model

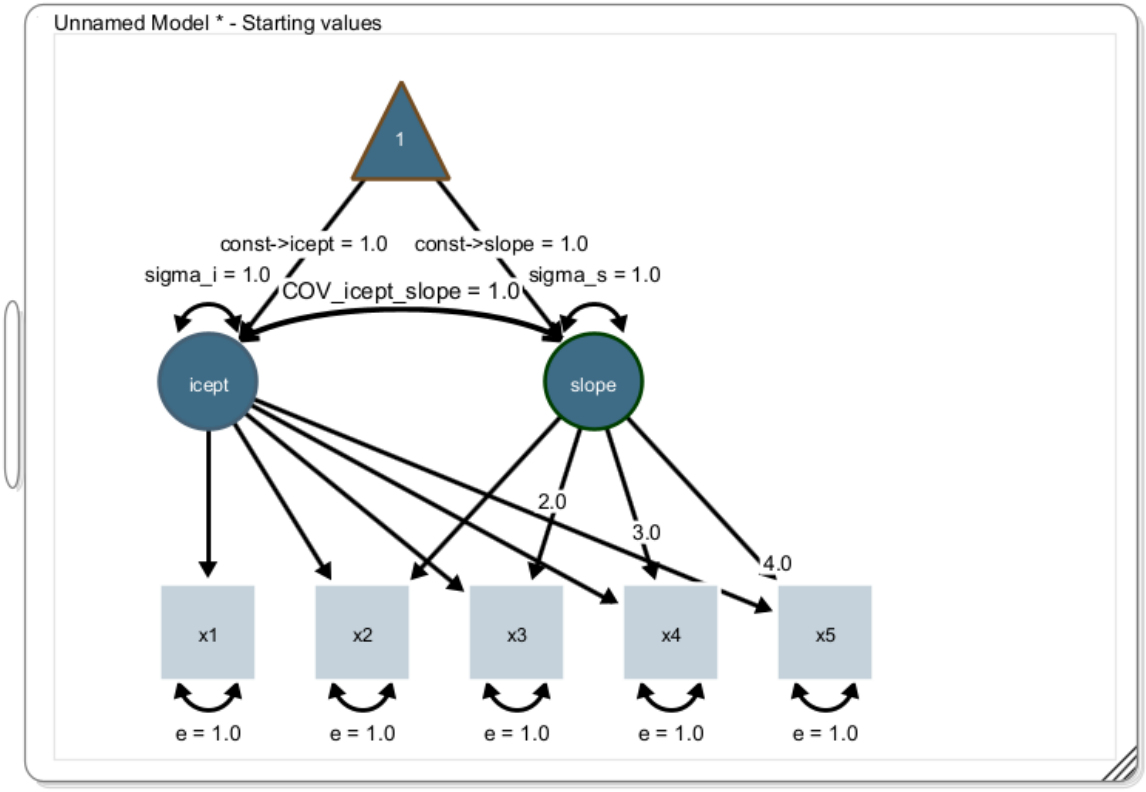
syntax is updated. Onyx highlights newly added code parts in yellow.

11.2 Exercise

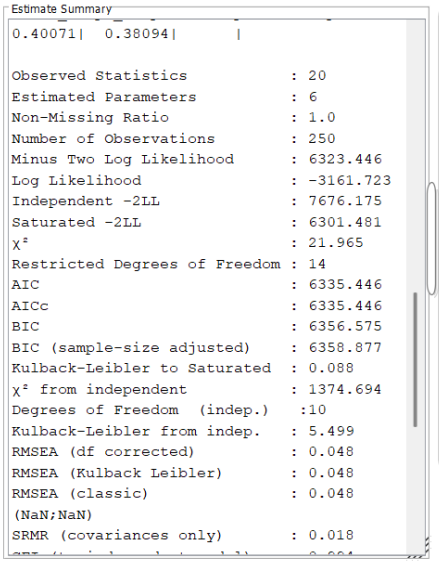
- Create a linear latent growth curve model using the linear latent growth curve model wizard (right-click the desktop, select "Create new model -> Create new LGCM"; make sure to estimate a covariance between intercept and slope (click the respective option in the dialog)
- Don't forget to use some of the visual options to change the appearance of the diagram
- Load the data "lgcm_simulated.csv" and obtain parameter estimates; inspect model fit
- Open the model syntax view for lavaan
- If you are familiar with lavaan and R: Copy the lavaan code to an R session; adapt the code to load the data file; estimate the model in lavaan; compare the results between Onyx and lavaan

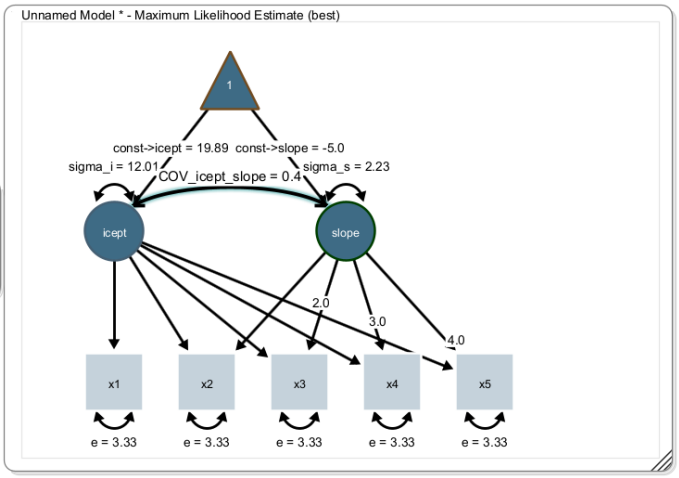
11.3 Solution

The path diagram should look like this:



Load the dataset, select all variables, right-click on the data view and click "send data to model". Model estimation starts and your parameter estimates and model fit statistics should look like this:






```
#
# This model specification was automatically generated by Onyx
#
library(lavaan);
modelData <- read.table(DATAFILENAME, header = TRUE) ;
 model<-"
   ! regressions
   icept=~1.0*x1
   icept=~1.0*x2
   slope=~1.0*x2
   icept=~1.0*x3
   slope=~2.0*x3
   icept=~1.0*x4
   slope=~3.0*x4
   icept=~1.0*x5
   slope=~4.0*x5
   ! residuals, variances and covariances
   x1 ~~ e*x1
   x2 ~~ e*x2
   x3 ~~ e*x3
   x4 ~~ e*x4
   x5 ~~ e*x5
   icept ~~ sigma_i*icept
   slope ~~ sigma_s*slope
```

```
   icept ~~ COV_icept_slope*slope
   ! means
   icept ~ const__icept*1
   slope ~ const__slope*1
   x1 ~ 0.0*1
   x2 ~ 0.0*1
   x3 ~ 0.0*1
   x4 ~ 0.0*1
   x5 ~ 0.0*1
"
result<-lavaan(model, data=modelData, fixed.x=FALSE, missing="FIML")
summary(result, fit.measures=TRUE)
```

In this code, you only need to replace the placeholder "DATAFILENAME" with the path to a data file on your computer. Then, model estimation in lavaan yields:

```
Warning: Paket 'lavaan' wurde unter R Version 4.5.3 erstellt


This is lavaan 0.6-21
lavaan is FREE software! Please report any bugs.


lavaan 0.6-21 ended normally after 52 iterations

  Estimator                                         ML
  Optimization method                           NLMINB
  Number of model parameters                        10
  Number of equality constraints                     4

  Number of observations                           250
  Number of missing patterns                         1

Model Test User Model:

  Test statistic                                21.965
  Degrees of freedom                                14
  P-value (Chi-square)                           0.079

Model Test Baseline Model:

  Test statistic                              1374.694
  Degrees of freedom                                10
```

```
  P-value                                       0.000

User Model versus Baseline Model:

  Comparative Fit Index (CFI)                   0.994
  Tucker-Lewis Index (TLI)                      0.996

  Robust Comparative Fit Index (CFI)            0.994
  Robust Tucker-Lewis Index (TLI)               0.996

Loglikelihood and Information Criteria:

  Loglikelihood user model (H0)             -3161.723
  Loglikelihood unrestricted model (H1)     -3150.741

  Akaike (AIC)                               6335.446
  Bayesian (BIC)                             6356.575
  Sample-size adjusted Bayesian (SABIC)      6337.554

Root Mean Square Error of Approximation:

  RMSEA                                         0.048
  90 Percent confidence interval - lower        0.000
  90 Percent confidence interval - upper        0.084
  P-value H_0: RMSEA <= 0.050                   0.497
  P-value H_0: RMSEA >= 0.080                   0.076

  Robust RMSEA                                  0.048
  90 Percent confidence interval - lower        0.000
  90 Percent confidence interval - upper        0.084
  P-value H_0: Robust RMSEA <= 0.050            0.497
  P-value H_0: Robust RMSEA >= 0.080            0.076

Standardized Root Mean Square Residual:

  SRMR                                          0.014

Parameter Estimates:

  Standard errors                             Standard
  Information                                 Observed
  Observed information based on                Hessian
```

```
Latent Variables:
                   Estimate  Std.Err  z-value  P(>|z|)
  icept =~
    x1                1.000
    x2                1.000
  slope =~
    x2                1.000
  icept =~
    x3                1.000
  slope =~
    x3                2.000
  icept =~
    x4                1.000
  slope =~
    x4                3.000
  icept =~
    x5                1.000
  slope =~
    x5                4.000

Covariances:
                   Estimate  Std.Err  z-value  P(>|z|)
  icept ~~
    slope   (COV_)    0.401    0.381    1.052    0.293

Intercepts:
                   Estimate  Std.Err  z-value  P(>|z|)
    icpt (cnst__c)   19.889    0.237   84.022    0.000
    slop (cnst__s)   -4.999    0.101  -49.371    0.000
   .x1                0.000
   .x2                0.000
   .x3                0.000
   .x4                0.000
   .x5                0.000

Variances:
                   Estimate  Std.Err  z-value  P(>|z|)
   .x1         (e)    3.335    0.172   19.365    0.000
   .x2         (e)    3.335    0.172   19.365    0.000
   .x3         (e)    3.335    0.172   19.365    0.000
   .x4         (e)    3.335    0.172   19.365    0.000
   .x5         (e)    3.335    0.172   19.365    0.000
    icept  (sigm_)   12.007    1.257    9.551    0.000
```

```
slope  (sgm_s)    2.230    0.230    9.699    0.000
```

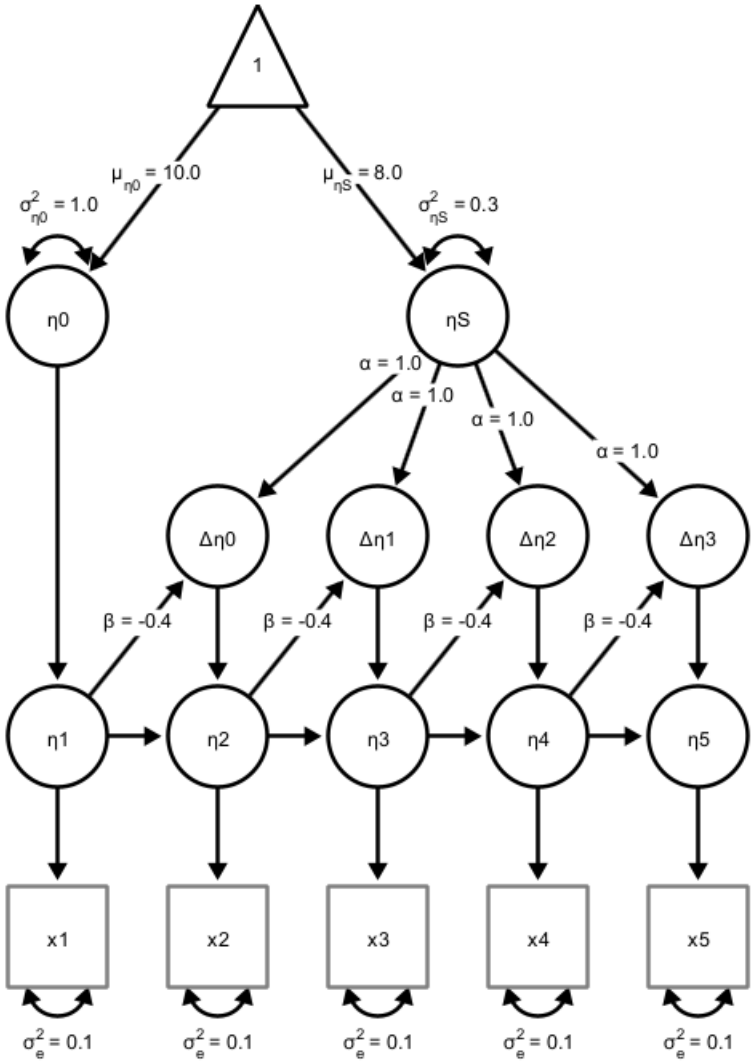
As you can see, all parameter estimates and fit statistics are identical.

12 Simulate and Plot

> Goal: Learn about simulation and plotting of data

12.1 The dual change score model

The dual change score model (DCSM) is a powerful framework within structural equation modeling (SEM) for analyzing longitudinal data by explicitly modeling how individuals change over time. Unlike traditional growth models that describe trajectories with fixed slopes, the DCSM decomposes change into two components: a constant (or proportional) change that depends on prior levels of the variable, and an additional dynamic component that captures how change at one time point influences subsequent change. This allows researchers to examine both stability and self-regulating processes within a construct, as well as potential coupling effects between multiple variables over time. By integrating features of latent growth curve models and autoregressive models, the DCSM provides a flexible approach for testing theories about developmental processes, feedback mechanisms, and temporal dependencies in psychological, behavioral, and social data. The model looks like this:



12.2 Simulation to Understand Model Parameters

In the remainder, we want to learn how Onyx simulation and plot features can be used. With these, we want to answer this question:

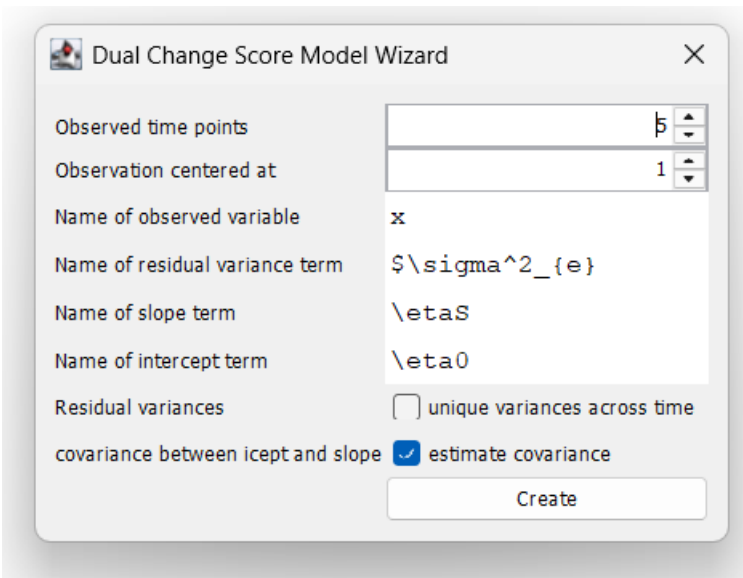
In a Dual Change Score Model, what effects do parameters have on the model-implied trajectories?

Steps:

1. Create a Dual Change Score Model
2. Simulate Longitudinal Data
3. Plot Longitudinal Data
4. Change Model (Population) Parameters and Repeat from 2

12.3 Dual Change Score Model Wizard

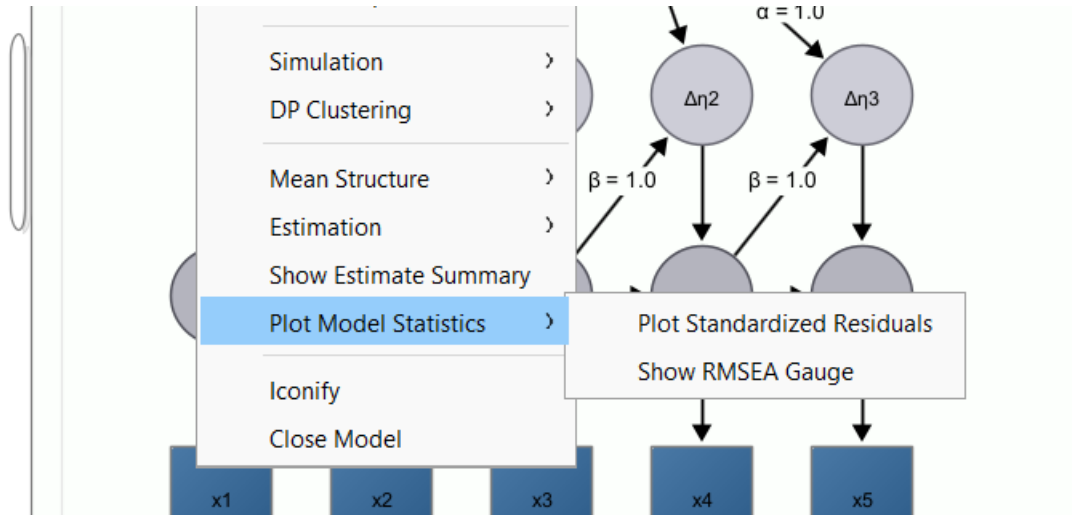
Use the DCSM-Wizard (right-click on the empty desktop, then choose "Create new model" and then select "Create new DCSM") to create a dual change score model. Keep all values at their defaults. This creates a model with five time points and the observed time points will be called x1, x2, ..., x5.



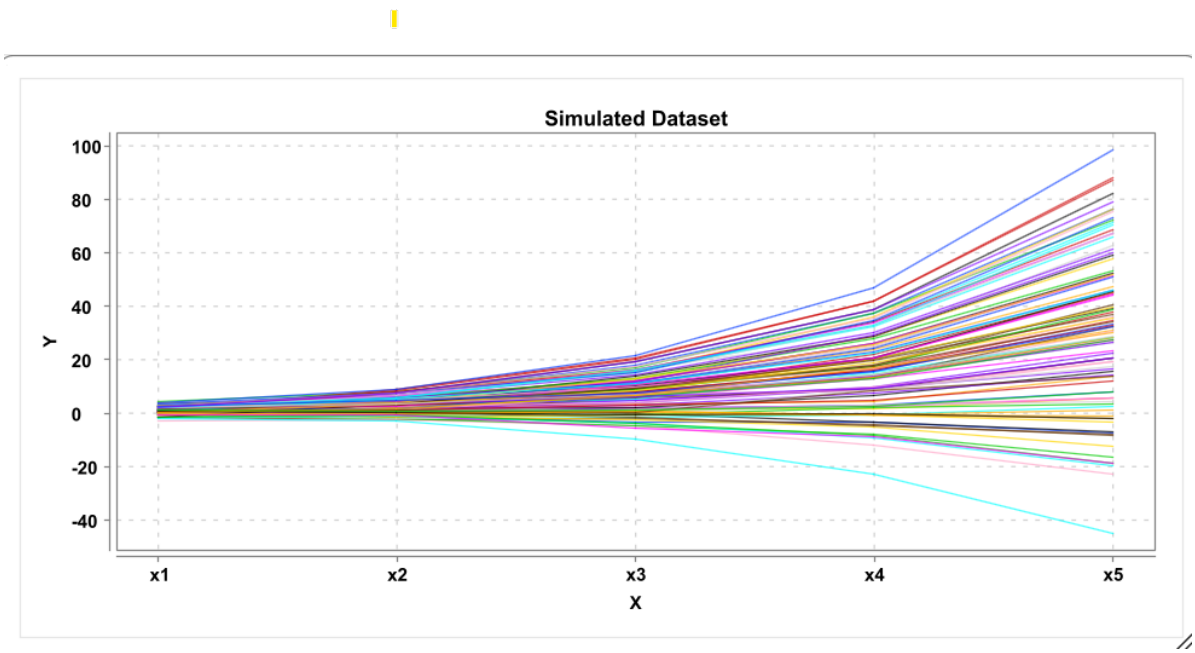
12.4 Simulation

12.5 Plot

Next, right-click on the simulated dataset, choose "Plot data" and select "Plot longitudinal data".



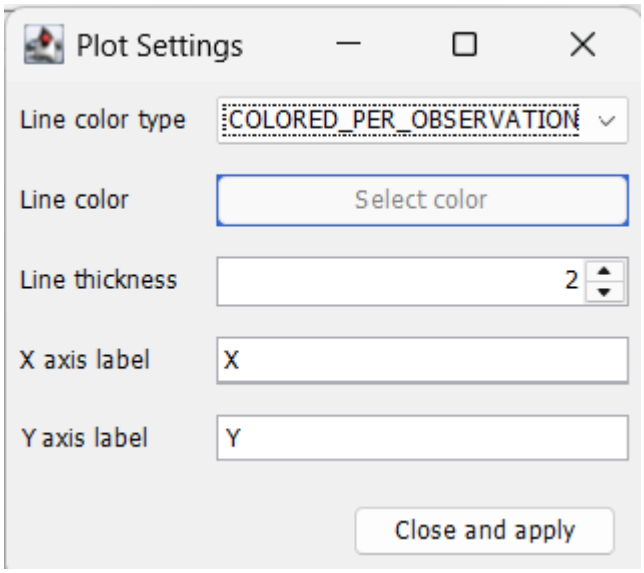
12.6 Longitudinal Plot



12.7 Customize the Longitudinal Plot

Right-click on the longitudinal plot window. This will show some basic plot settings. Try out different ways to customize the plot. By default, the line color is selected as "COL-

ORED_PER_OBSERVATION" such that each observation gets a different color. Alternatively, you can set the color to "USER DEFINED" and then pick a line color. You can also change the line thickness, and the axes labels



Click "close and apply" to apply the settings and close the dialog window. Right-click again and choose "save PNG" to save the graph as image file.

12.8 Plot subsets of variables

If no variables are selected in the dataset, the longitudinal plot will be plotted across all variables in the dataset (in the order, in which they appear in the dataset). If you want to plot only across a subset of variables, select those variables in the variable list (e.g., by holding CTRL or CMD down and select the respective variables) and then click "Plot Data -> Plot longitudinal data".

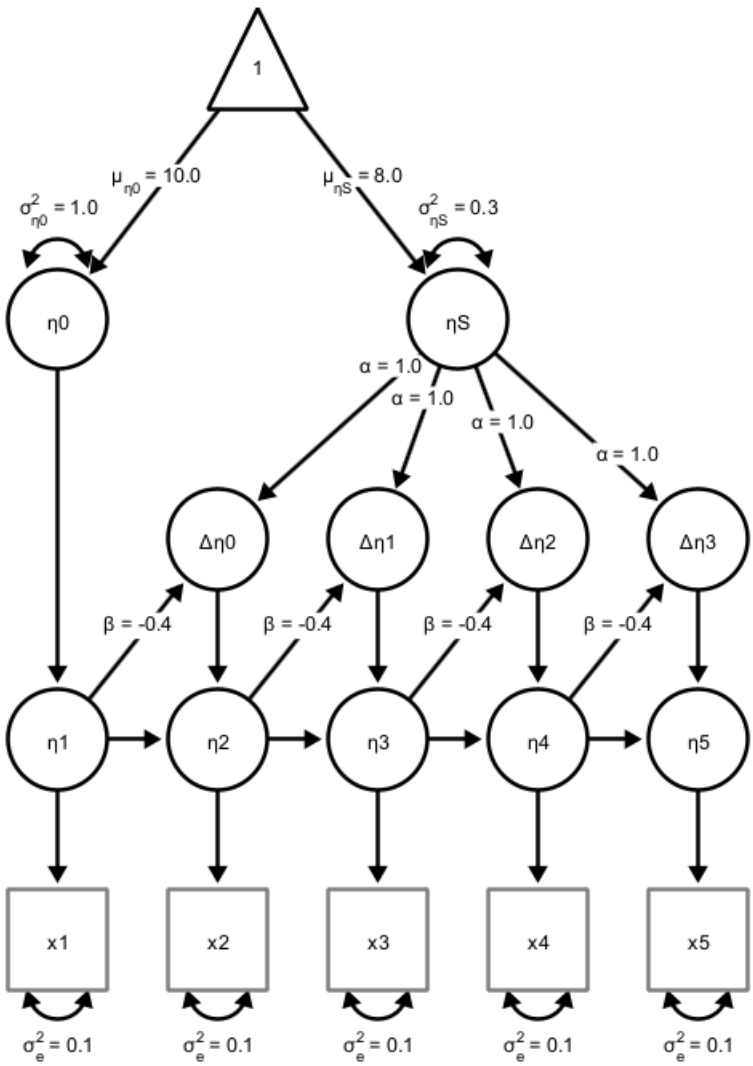
12.9 Exercises

- Create a dual change score model using the wizard
- Simulate Data and plot the data (longitudinal plot)
- Change Parameters of the model, resimulate and plot
- What happens if the self-feedback (beta) is 1, 0, or -1? Why?

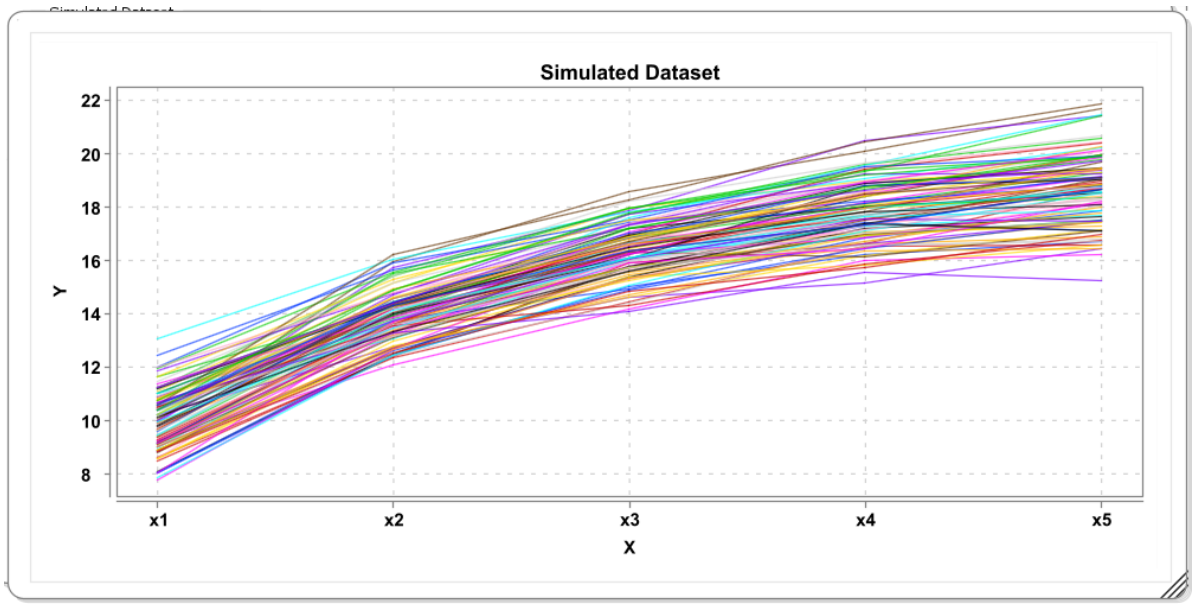
- If you set the self-feedback to zero, what are the effects of changing alpha? What if you set different alphas (rename them to alpha1, alpha2,… first) for different time points?
- By setting a form of constant positive change (alpha) and negative proportional change (beta), can you generate simulated learning curves that reach an asymptote?

12.10 Solution

The self-feedback term beta implies a form of proportional growth if values are positive because change then depends on previous level. This by itself can reflect an exponential growth function. In combination with constant growth rates (alpha > 0), negative proportional change can be used to model change that flattens after a while. Here is one possible solution:



And these are some simulated trajectories from the model (N=100):


Simulated Dataset
22
20
18
16
14
12
10
8
Y
x1
x2
x3
x4
x5
X

13 Exercises

In the following chapters, you will find exercises to practice your Onyx and modeling skills.

14 Exercise: Multiple determinants of lifespan memory differences

In this exercise, we will reproduce results reported by (Henson et al. 2016). In their study, they speculate on different theories of memory. Here, we use SEM to test different measurement models representing the different theories of memory. Items are 9 memory scores (3 valences and 3 tests, fully crossed). The tests tap into associative memory, item memory, and visual priming. Valences are positive, neutral, negative. The data is based on the reported covariances of the item d' values across the nine tests. Henson et al. (2016) write in their introduction:

> Human memory is often claimed to consist of different systems, supported by distinct brain regions[1],[2],[3]. However, the precise fractionation of these memory systems is a matter of continuing debate[4],[5].

In the following, they list a couple of different theories that postulate different structural models underlying the observed variances and covariances.

14.1 Theories

Alternative theories for memory discussed are:

- Theory #1 (Berry et al.)

A single factor drives memory performance across all tests and valences.

- Theory #2 (Squire)

Two components drive memory performance, namely declarative (explicit) memory underlying associate and item memory, plus a procedural (implicit) system underlying priming.

- Theory #3 (Gardiner et al., Yonelinas)

An alternative two-component model assumes a recollection factor underlies associate memory, and a fluency factor underlies item memory and priming.

(optional variant: item memory is driven by both recollection and fluency)

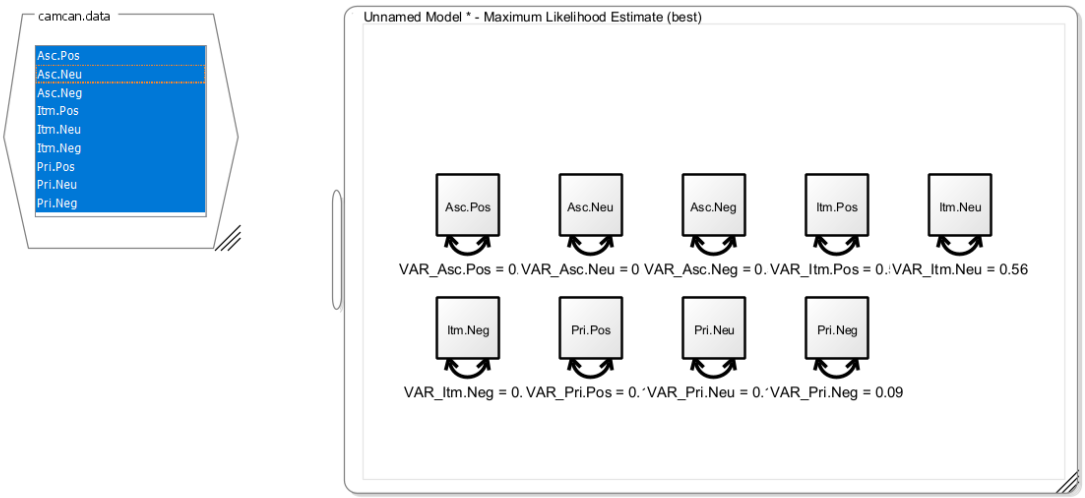
- Theory #4

A three-component model predicts that each test is best described by an individual underlying component.

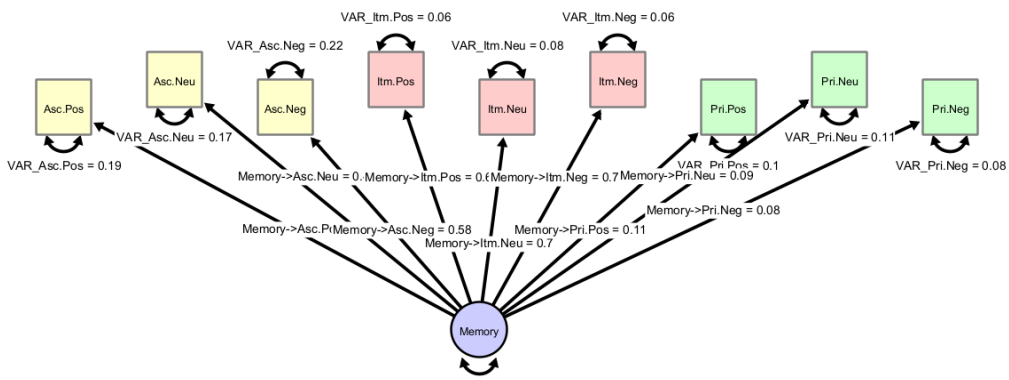
14.2 Fast Model Creation

The data are provided in the file `camcan.data`, which contains the variances and covariances across the nine indicators measured in N=305 persons.

Note that the fastest way to start modeling is to load the data, select all variables in the loaded dataset and drag them onto the empty model panel. This way, all observed variables are created in the model, together with each a free parameter representing their variance, and they are directly linked to the dataset (for model fitting). Maximum likelihood-estimation directly starts and the values represent the maximum-likelihood estimates of the variances of each variable.



In order, to create a common factor, double-click on an empty space in the model view. A latent variable (shown as a circle) appears. Change the default name to "Memory". Now add the factor loadings. These are directed paths from the latent factor to each observed variable and make them freely estimated variables. Feel free to use the layout and graph settings to customize the appearance. Note that a factor model needs an identification constraint to identify the latent scale. To implement this, either fix the latent variance to one or fix one of the loadings to one. In the first case, the latent scale is identified such that it has unit variance; in the second case, it is identified to have the same unit of measurement as the item with the unit factor loading. Your graph should look somewhat like this:



Let us inspect the explained variance in each item. You can do this visually by giving the observed variables a fill color and then select "Customize Variable -> Select Fill Style -> Bokers R2" and each observed variable will be filled according to the level of explained variance. Or, you hover the mouse over a single, unselected observed variable. A popup window will appear, in which you can read off the R^2 value as estimate of explained variance. Using either way, we can already see that the priming variables are not well explained by the common factor. Further, we find these goodness-of-fit values:

```
RMSEA (classic)                : 0.288
SRMR (covariances only)        : 0.142
CFI (to independent model)     : 0.73
```

What other models could explain these data better?

14.3 Exercises

Exercise:

1) Set up models representing the alternative theories, fit the models with the provided data (camcan.data) as described above and discuss the results.

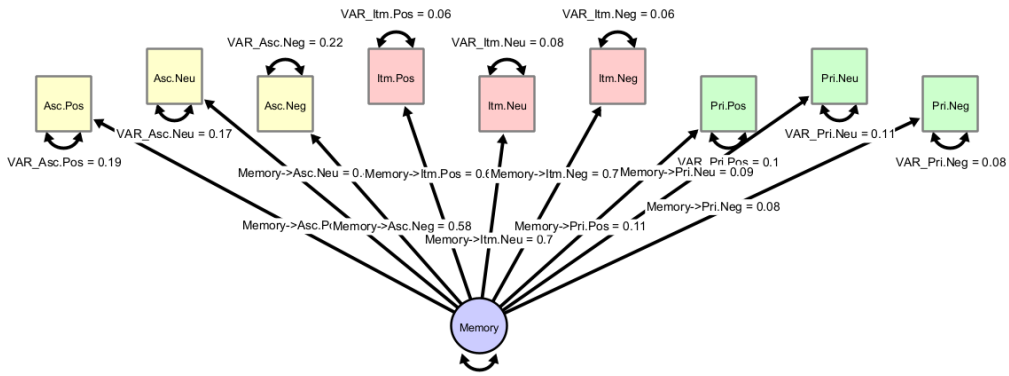
2) Inspect parameter estimates (e.g., check for problems like negative variances) and model fit statistics

3) What model fits best?

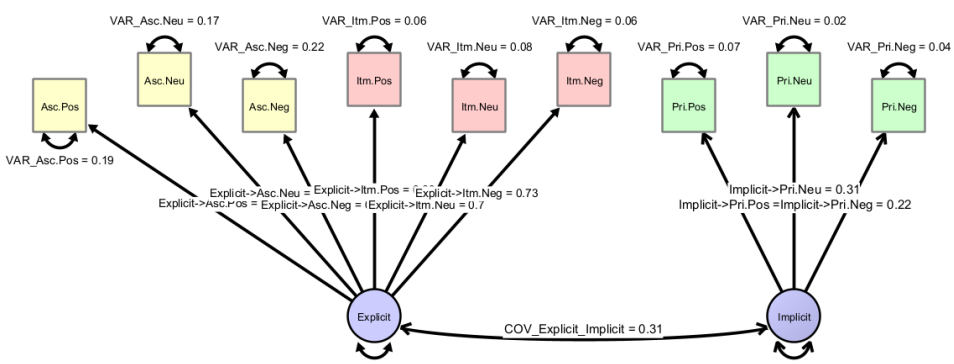
4) How could your final model incorporate predictors of behavior, e.g. grey matter volume estimates of some ROIs?

14.4 Solution

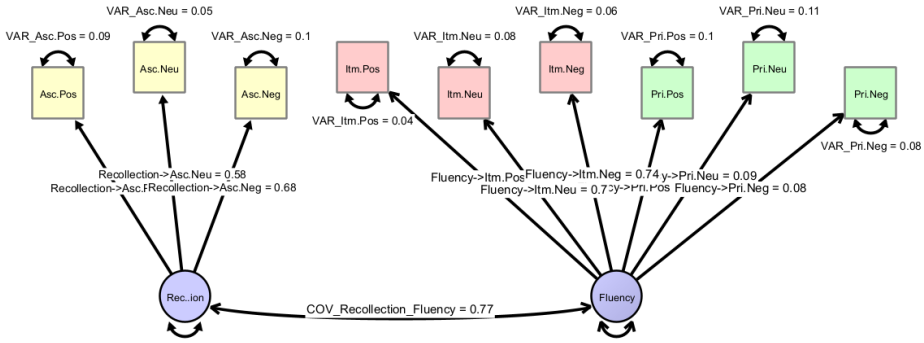
This is the common factor model (Theory 1):



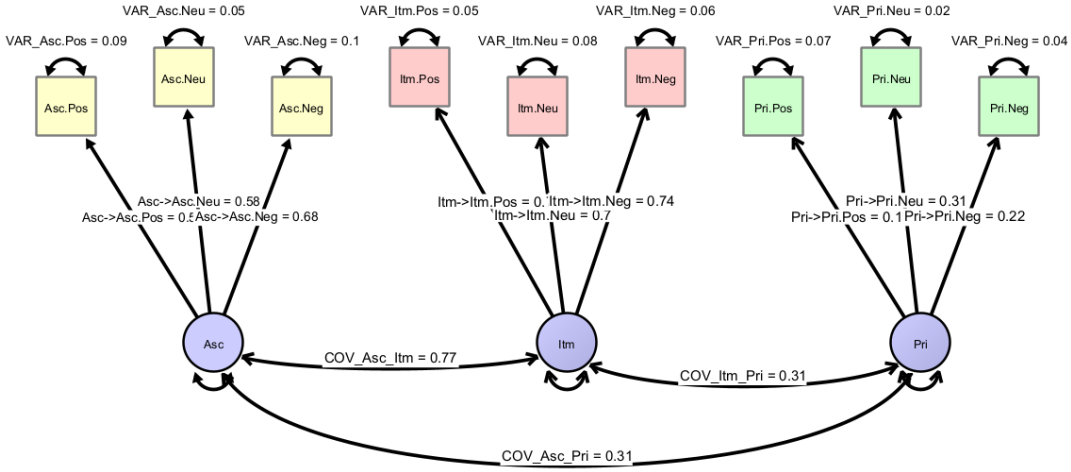
This is explicit and implicit memory (Theory 2):



This is the recollection and fluency factor (Theory 3):



And finally a model of three different factors:

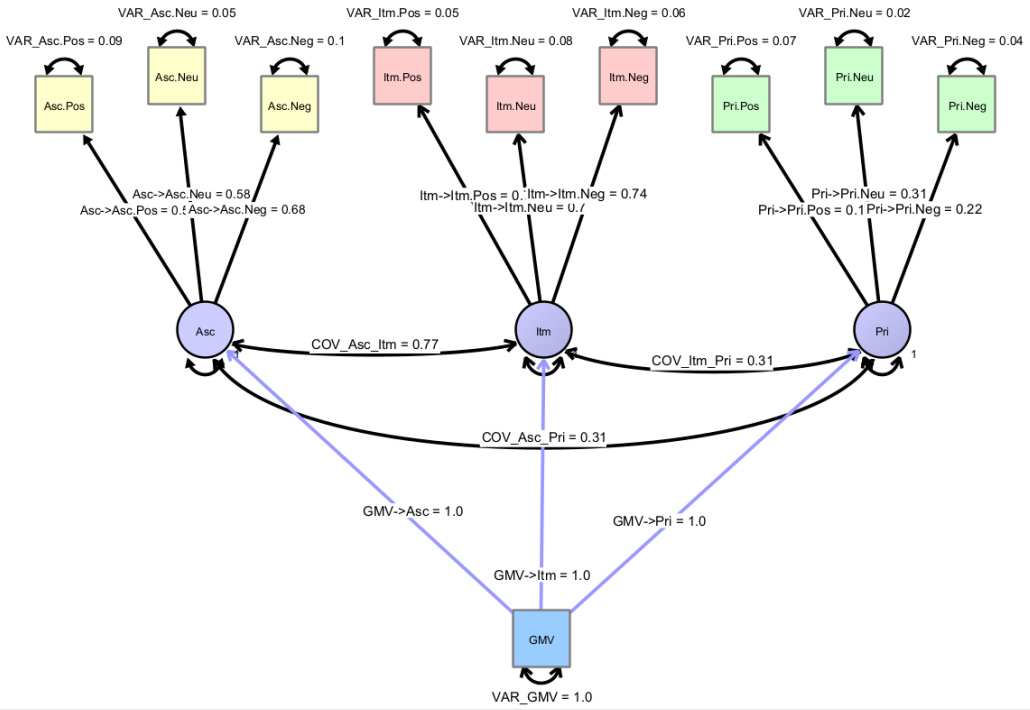


14.4.1 Model Comparison

For all models, inspect model fit (CFI, RMSEA, residuals) and also investigate the factor loading strengths as well as the explained variances across items.

14.4.2 Extension to Brain Variables

How would a model look like, which has a brain variable, which we would like to use to predict behavior with?


VAR_Asc.Neu = 0.05
VAR_Itm.Pos = 0.05
VAR_Itm.Neg = 0.06
VAR_Pri.Neu = 0.02
VAR_Asc.Pos = 0.09
VAR_Asc.Neg = 0.1
VAR_Itm.Neu = 0.08
VAR_Pri.Pos = 0.07
VAR_Pri.Neg = 0.04
Asc.Pos
Asc.Neu
Asc.Neg
Itm.Pos
Itm.Neu
Itm.Neg
Pri.Pos
Pri.Neu
Pri.Neg
Asc->Asc.Neu = 0.58
Asc->Asc.Neg = 0.68
Itm->Itm.Neg = 0.74
Pri->Pri.Neu = 0.31
Pri->Pri.Neg = 0.22
Asc
Itm
Pri
COV_Asc_Itm = 0.77
COV_Itm_Pri = 0.31
COV_Asc_Pri = 0.31
GMV->Asc = 1.0
GMV->Pri = 1.0
GMV->Itm = 1.0
GMV
VAR_GMV = 1.0

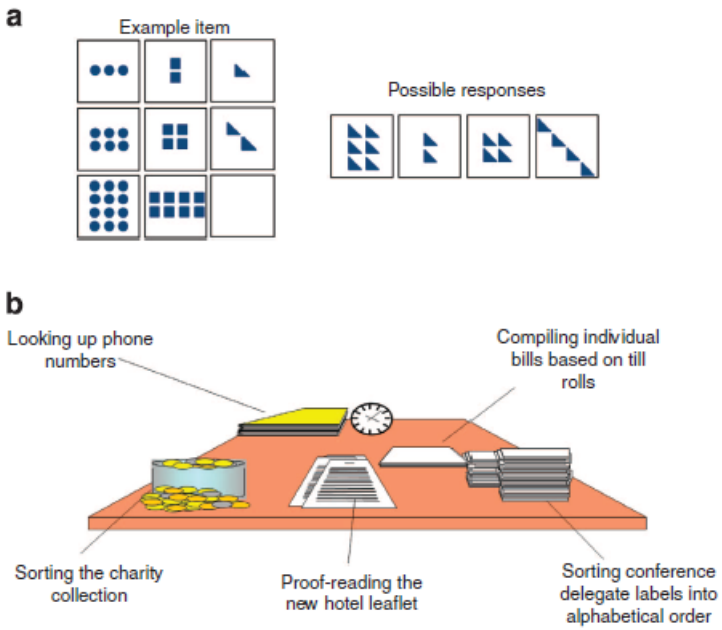
15 Age-related differences in in fluid intelligence and multitasking and their correlations with gray-matter and white-matter volume

In this exercise, we will reproduce an published SEM analysis by Kievit et al. (2014). From their study:

> We focus on four neural properties selected based on the current literature—(see Methods), two involving grey matter volume (GMV; Brodmann Area 10 (BA10) and the Multiple Demand (MD) System), and two involving WMI (the Forceps Minor (FM) and the Anterior Thalamic Radiations (ATR)). Fitting a series of structural equation models, we show that multitasking and fluid intelligence are distinct cognitive abilities that show diverging age-related differences.

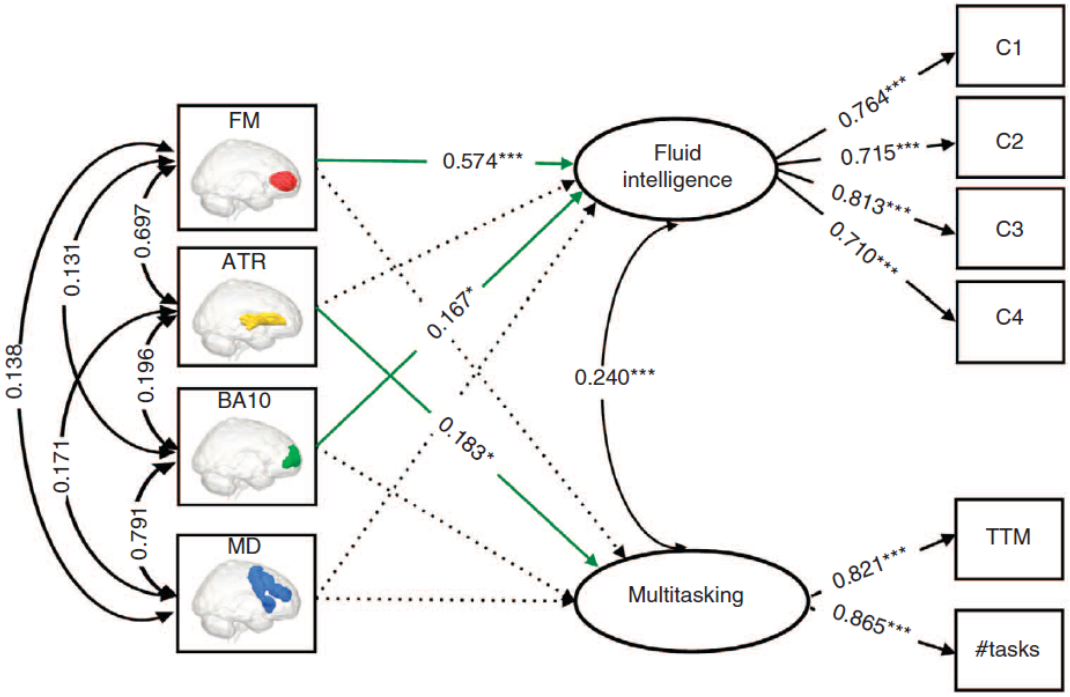
In their study, they assessed two behavioral performance measures. From their study:

> We assess fluid intelligence using the Cattell Culture Fair Test, consisting of pencil-and-paper subtests that yield four summary scores (series completions, odd-one-out, matrices and topology) used in further modelling. […] This consisted of four subtests yielding a sum score each. In contrast to the Cattell test of fluid intelligence, the Hotel test simulates a hotel work environment and measures the ability to distribute performance across multiple tasks, which we will refer to from here on as multitasking.



> Ageing is characterized by declines on a variety of cognitive measures. These declines are often attributed to a general, unitary underlying cause, such as a reduction in executive function owing to atrophy of the prefrontal cortex. However, age-related changes are likely multifactorial, and the relationship between neural changes and cognitive measures is not well-understood. Here we address this in a large (N=567), population-based sample drawn from the Cambridge Centre for Ageing and Neuroscience (Cam-CAN) data. We relate fluid intelligence and multitasking to multiple brain measures, including grey matter in various prefrontal regions and white matter integrity connecting those regions.

Our final goal is to reproduce their final SEM to explain individual differences in the two cognitive facets using the four brain measures of interest:

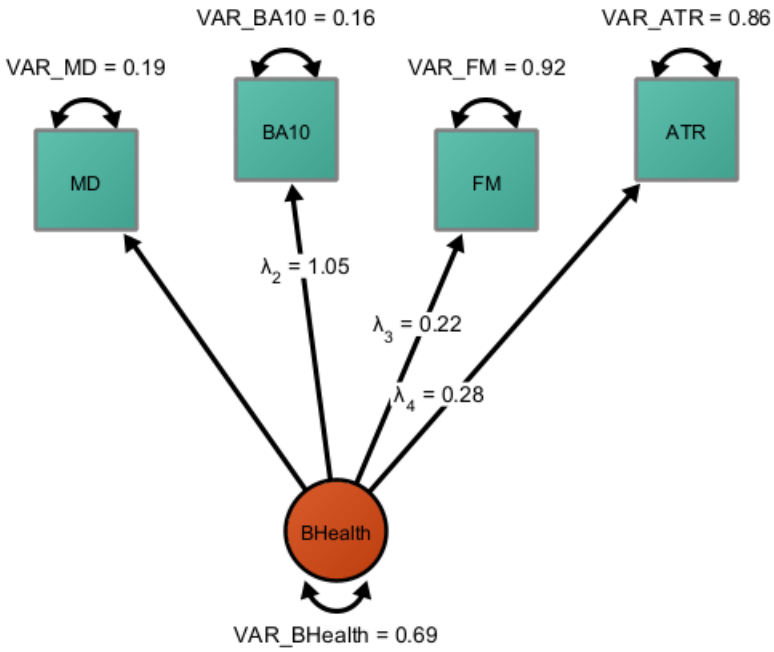

FM
ATR
BA10
MD
0.574***
0.167*
0.183*
0.240***
Fluid intelligence
Multitasking
0.764***
0.715***
0.813***
0.710***
0.821***
0.865***
C1
C2
C3
C4
TTM
#tasks
0.138
0.131
0.697
0.196
0.171
0.791

16 Exercise

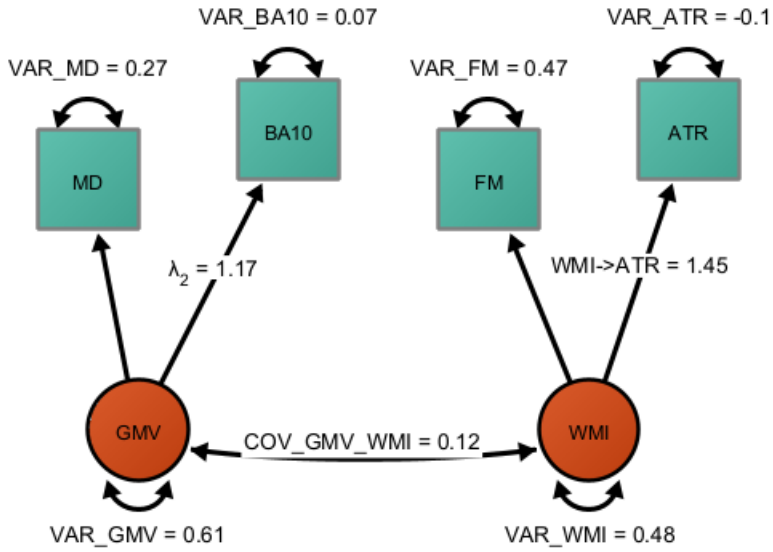
1. Load the dataset "ncomms.dat", which contains the variances and covariances as reported by Kievit et al. (2014).
2. Create a measurement model for cognition using only the behavioral variables. What model fits better? A model with a single performance factor across all six tests? Or two correlated latent factors, one fluid intelligence factor for the Cattell items and one multitasking factor for the Hotel test items?
3. Create a measurement model for the brain? Does a single factor across gray matter volume and white matter volume represent the observed data well? If not, is it possible to represent the data using two correlated factors? A gray matter volume factor with two indicators and a white matter factor with two indicators? If not, what else can we do?
4. Fuse both models in one model and use whatever items or factors you chose in your measurement models, such that the brain variables predict behavior. Freely estimate those parameters. What can we learn?
5. Discuss the results (e.g., directionality and size of the effects)
6. Test whether the effect of FM->MT and ATR->MT is different.

16.1 Solution

Here are some alternative model for the brain variables. First, we model a single factor:



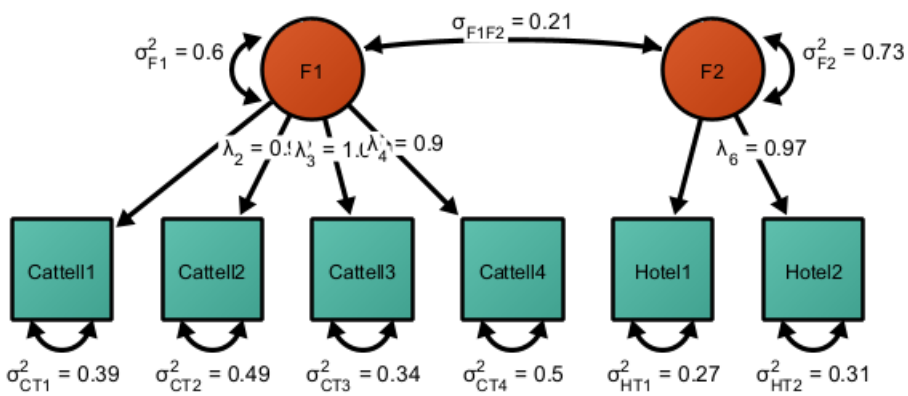
However, model fit and explained variances are not satisfying. Let us try a factor model per modality:



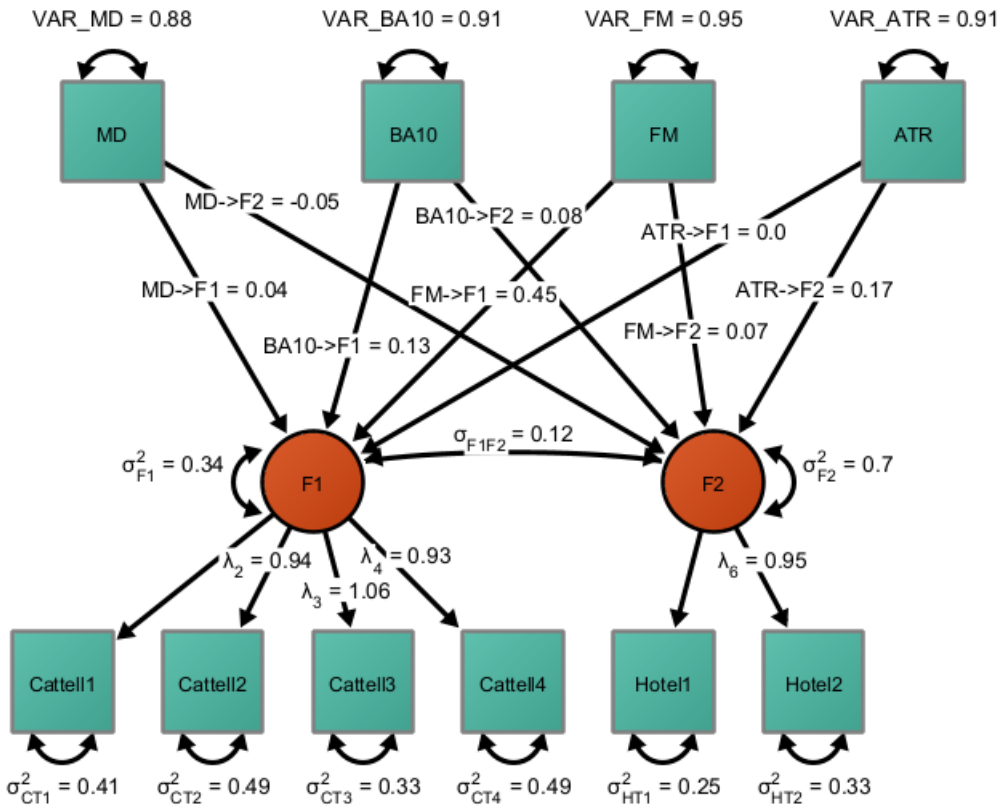
However, a negative variance is an indicator for an unsatisfying solution. So, we will not

estimate any latent factors and instead keep the observed variables.

Here is a measurement model for the behavioral variables (how does this compare to a single-factor model?):



Here is the final model:


VAR_MD = 0.88
VAR_BA10 = 0.91
VAR_FM = 0.95
VAR_ATR = 0.91
MD
BA10
FM
ATR
MD->F2 = -0.05
BA10->F2 = 0.08
ATR->F1 = 0.0
MD->F1 = 0.04
FM->F1 = 0.45
ATR->F2 = 0.17
BA10->F1 = 0.13
FM->F2 = 0.07
σ F1F2 = 0.12
σ2 F1 = 0.34
σ2 F2 = 0.7
F1
F2
λ2 = 0.94
λ4 = 0.93
λ3 = 1.06
λ6 = 0.95
Cattell1
Cattell2
Cattell3
Cattell4
Hotel1
Hotel2
σ2 CT1 = 0.41
σ2 CT2 = 0.49
σ2 CT3 = 0.33
σ2 CT4 = 0.49
σ2 HT1 = 0.25
σ2 HT2 = 0.33

17 The End

You have reached the end of the workshop. Well done!

Please let us know whether the workshop was helpful. Let us know if you encountered any problems, if things were not clear, or if you just were happy with how things were. We are always happy about any form of constructive feedback to improve our workshops and our software.

Feel free to get in touch with Andreas: brandmaier@mpib-berlin.mpg.de

References

Boker, Steven M, Timo von Oertzen, Joshua N Pritikin, Michael D Hunter, Timothy R Brick, Andreas M Brandmaier, and Michael C Neale. 2023. "Products of Variables in Structural Equation Models." *Structural Equation Modeling: A Multidisciplinary Journal* 30 (5): 708–18.

DeShon, Richard P. 1998. "A Cautionary Note on Measurement Error Corrections in Structural Equation Models." *Psychological Methods* 3 (4): 412–23.

Hancock, Gregory R. 2001. "Effect Size, Power, and Sample Size Determination for Structured Means Modeling and MIMIC Approaches to Between-Groups Hypothesis Testing of Means on a Single Latent Construct." *Psychometrika* 66 (3): 373–88.

Henson, Richard N, Karen L Campbell, Simon W Davis, Jason R Taylor, Tina Emery, Sharon Erzinclioglu, and Rogier A Kievit. 2016. "Multiple Determinants of Lifespan Memory Differences." *Scientific Reports* 6 (1): 32527.

Kievit, Rogier A, Simon W Davis, Daniel J Mitchell, Jason R Taylor, John Duncan, and Richard NA Henson. 2014. "Distinct Aspects of Frontal Lobe Structure Mediate Age-Related Differences in Fluid Intelligence and Multitasking." *Nature Communications* 5 (1): 5658.